Drawing register and engineering basis

Controlled source snapshot; all physical dimensions are millimetres unless marked otherwise.

DESIGN STATUS

1. This book coordinates the current Johansson World ferry with explicitly identified reference machinery and proposed reservations. It is a review issue, not an approved real-vessel construction package.

2. Existing source geometry is an audit baseline. Proposed reservations are shown separately; they are not automatically substituted into the game hull.

3. Drawing layout uses fixed ISO A paper sizes, borders, revision identification, drawing numbers, datums, units and a title block. It does not assert full compliance with any classification standard.

4. No signature, class stamp, flag approval, manufacturer certification or professional registration is implied.

5. Construction release holds include scantlings, foundations, stability and damage stability, fire boundaries, machinery vendor installation data, hydraulic sizing and electrical fault/selectivity calculations.

Particular	Source / basis
LOA / LBP / moulded breadth	34.5 / 31.0 / 8.4 m
Moulded depth / design draft	3.0 / 1.9 m
Machinery floor / deck underside	900 / 2800 mm above BL
Nominal clear room height	1900 mm before pipes and trays
Engine-room longitudinal limits	8500 - 14000 mm from transom
Auxiliary / workshop limits	14000 - 19000 / 19000 - 22500 mm
Internal side reservation	Y = +/- 3980 mm
Main-engine game power	2 x 460 kW at 1350 rpm
Nominal AC main system	440 V, 3 phase, 60 Hz, IT
CAD revision / issue date	B /2026-10-13

READING THE DRAWINGS

1. X increases forward from the transom. Y increases to port from the centreline. Z increases upward from the baseline.

2. Game [x, y, z] metres -> CAD [1000(z + LOA/2), 1000x, 1000(y + draft)] millimetres.

3. Geometry DXFs are model space at 1:1 in mm. PDF view scale is stated individually. Schematic DXFs are diagram coordinates and are NTS.

4. Source snapshot, node/line schedules, equipment bounds and cable schedules are supplied in data/. See README for reproduction and plotting.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

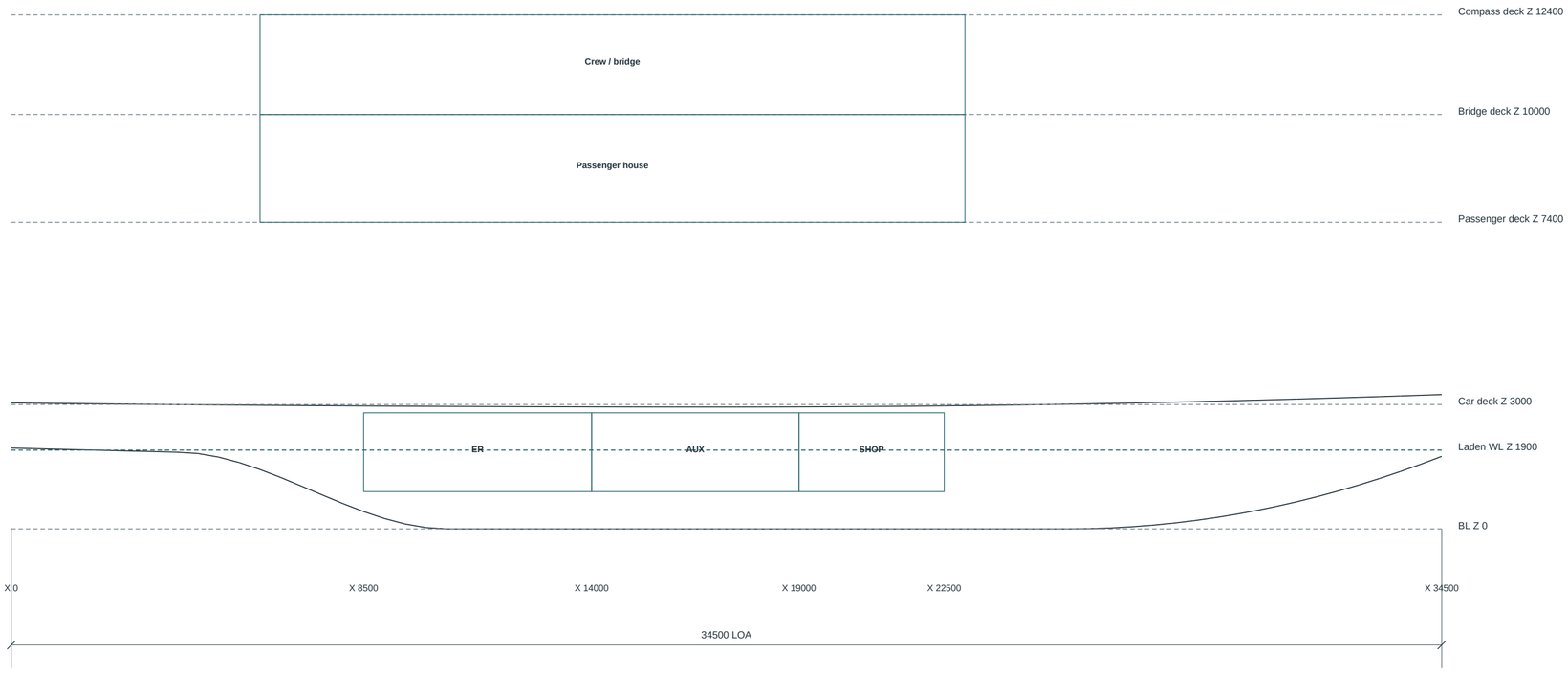
JS127-ER-000 | Drawing register and engineering basis

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 1

Machinery spaces in vessel context

Profile and deck datums from the current game loft; weight and stability remain unreconciled.



CONNECTION TO THE GAME

1. Length34500, breadth8400, depth3000 mm. These are source design dimensions.
2. Each equipment envelope and pipe centre-line converts through a single documented axis transform.
3. Corrections to source allocations appear in the following arrangement sheets; overhaul, escape and regulatory holds remain explicit.

CONTEXT AND HYDROSTATIC QUALIFICATIONS

1. The hull loft at Z 1900 draft yields approximately 375.4 t geometric displacement. This is not verified lightship or certified payload capacity.
2. Legacy operating weight target283 t and lightship210 t remain unverified. An arithmetic difference between hull displacement and assumed lightship is not an approved allowable deadweight.
3. The existing shallow machinery depth is retained; a deeper hull or raised deck/casing would alter hydrostatics, stability, car-deck interfaces and structural design.
4. Profile shows compartment/datums only. No plate thickness, deck-beam section, longitudinal stiffness or approved subdivision is established.

Scale bar: 5 m at 1:100 (100% print)



Source e87c06b16734 | CAD units mm | dimensions govern
Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU
FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-100 | Machinery spaces in vessel context

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:100 | SHEET 2

Main machinery and propulsion particulars

Johansson equipment particulars are fictional design assumptions. Sourced modern reference is a separate comparison.

Particular	Current game / derived value	Status
Main engine / manufacturer	2 x Johansson JM6-160	Fictional
Engine type	Four-stroke medium-speed marine diesel, six in line, turbocharged and...	Design assumption
Rated brake output	460 kW each at 1350 rpm	Unverified maker data
Cylinders / bore / stroke	6 in line /160 /200 mm	Unverified maker data
Displacement / piston speed	24.127 L /9.00 m/s	Derived from assumptions
Gear type / ratio	Reverse reduction /3.00:1	Vendor selection unresolved
Shaft diameter / rated shaft speed	110 mm /450 rpm	Sizing and torsion unverified
Ideal torque after gearbox	9.762 kNm per shaft (no losses)	Not shaft approval
Propeller / diameter / pitch ratio	2 x 4 blades /1450 mm /0.8	Resistance/cavitation unverified
Main service pumps	Engine-driven LO, JW and sea-water	Maker capacity unresolved
Starting system	Air start; 2 x 150 L bottles; nominal 30 bar	Starting energy/maker verification open
Specific fuel consumption / density	210 g/kWh /0.86 kg/L	Assumed duty calculation inputs
Crossing fuel /20 crossing day	73.10 L /1462.1 L	80%main load 22 min + assumed generator
Fuel storage /service tank	15.0 m3 /1.5 m3	Usable capacity/structure open
Yanmar comparison rating	6AYM-WST485 kW/1900 rpm/20.379 L	Published reference; separate duty
Published reference engine outline	2000 x 1305 x 1431 mm; 2365 kg dry	Legacy family catalogue outline
Matched gear /overall length	YX181; 3.03:1; 560 kg; combinedL2364 mm	Reference family; exact install open
Yard proposed combined allowance	2700 x 1500 x 1650 mm; 3.6 t each	Reservation, not certified assembly

INSTALLATION RELEASE DATA STILL REQUIRED

1. Selected engine and gearbox installation sheets: mounting holes, resilient mounts or chocks, bedplate loads, nozzle coordinates, start/control interfaces and cooling duties.
2. Shaft alignment, bearing spans/reactions, propeller thrust, fatigue and torsional vibration; exhaust backpressure and ventilation; certified lifting points and service/removal procedures.
3. Reference dimensions and rating may be from different published family duties. They must not be combined into an approved selected-machine specification without vendor confirmation.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

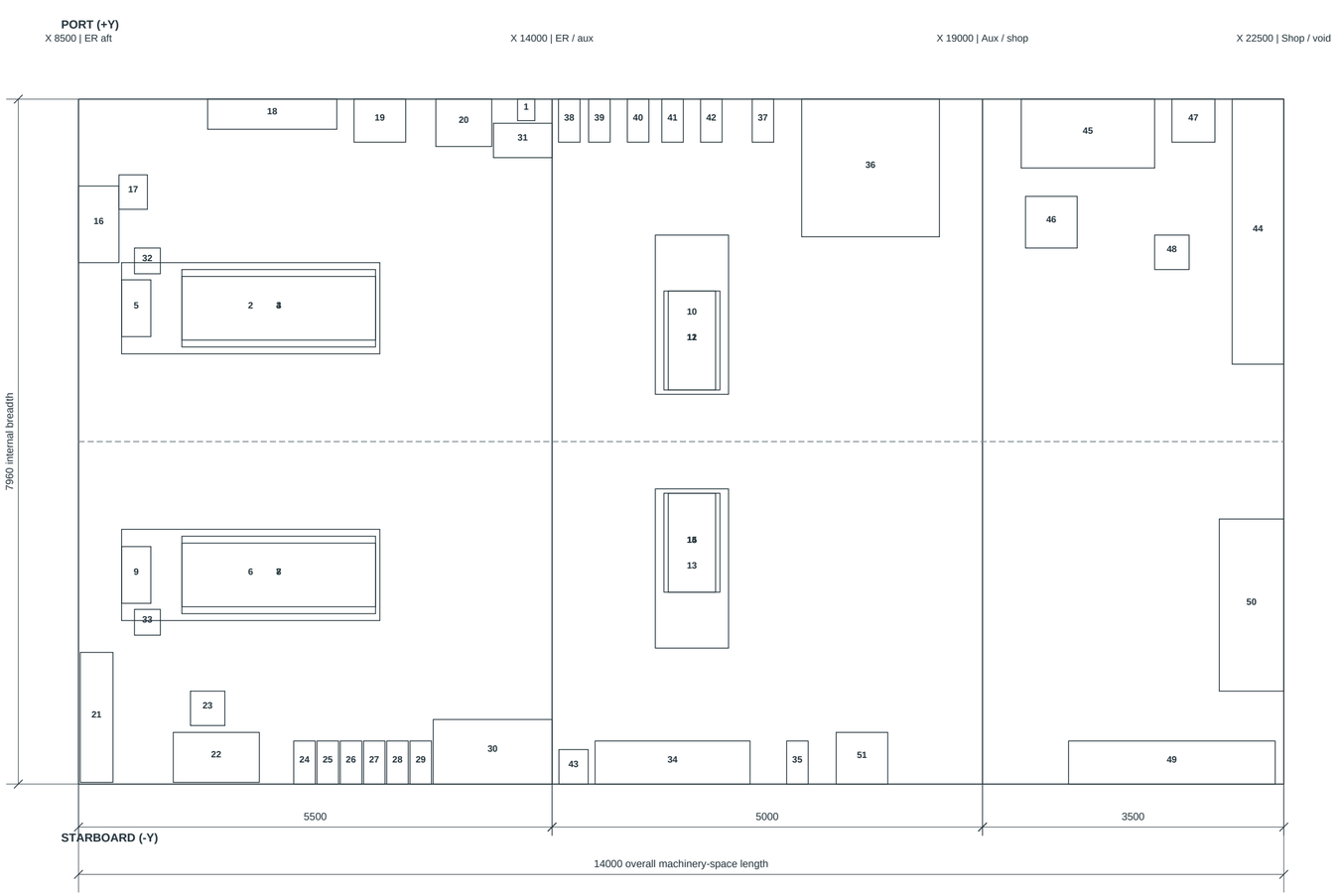
JS127-ME-001 | Main machinery and propulsion particulars

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 3

Machinery spaces - corrected source arrangement

Plan at floor plates Z = 900 mm; includes source corrections and retained unresolved engineering holds.



Item	Model equipment / object	X limits	Y limits
1	hydrant	13600-13800	3730-3980
2	main engine bedplate	9000-12000	1020-2080
3	main engine sump	9700-11950	1180-1920
4	main engine	9700-11950	1100-2000
5	gearbox	9000-9340	1220-1880
6	main engine bedplate	9000-12000	-2080--1020
7	main engine sump	9700-11950	-1920--1180
8	main engine	9700-11950	-2000--1100
9	gearbox	9000-9340	-1880--1220
10	generator bed	15200-16050	550-2400
11	generator sump	15350-15900	600-1750
12	generator engine	15300-15950	600-1750
13	generator bed	15200-16050	-2400--550
14	generator sump	15350-15900	-1750--600
15	generator engine	15300-15950	-1750--600
16	air bottles	8500-8970	2080-2970
17	air compressor	8970-9300	2700-3100
18	workbench	10000-11500	3630-3980
19	fuel purifier	11700-12300	3480-3980
20	oily-water separator	12650-13300	3430-3980
21	main switchboard	8520-8900	-3960--2450
22	engine panel and log desk	9600-10600	-3960--3380
23	booth chair	9800-10200	-3300--2900
24	pump	11000-11250	-3980--3480
25	pump	11270-11520	-3980--3480
26	pump	11540-11790	-3980--3480
27	pump	11810-12060	-3980--3480
28	pump	12080-12330	-3980--3480
29	pump	12350-12600	-3980--3480
30	fuel service tank	12620-14000	-3980--3230
31	sea-water strainer	13320-14000	3300-3700
32	local control stand	9150-9450	1950-2250
33	local control stand	9150-9450	-2250--1950
34	ballast manifold	14500-16300	-3980--3480

17 further objects in equipment_geometry.csv.

REVIEW FINDINGS

1. Current engine symbols, gears, generators and switchboard are game geometry. No vendor bolt pattern, vibration allowance or lifting attachment is established.

2. Room height is only 1900 mm. Pipes/trays further reduce clearance; piston extraction cannot be approved from this arrangement.

3. Numbered silhouettes are source objects, including collision reservations. Complete bounds and original source indices are in equipment_geometry.csv.

Scale bar: 5 m at 1:50 (100% print)

0 1 2 3 4 5

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-101 | Machinery spaces - corrected source arrangement

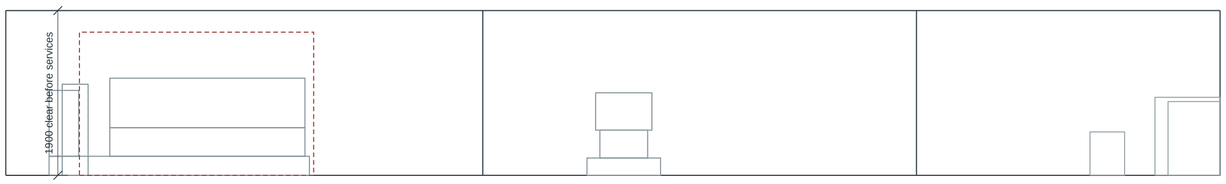
DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 4

Machinery spaces - longitudinal and transverse sections

Sections use current floor and deck levels; transverse machinery shown as combined envelope projection.

A-A LONGITUDINAL SECTION / PROJECTED MACHINERY ENVELOPES

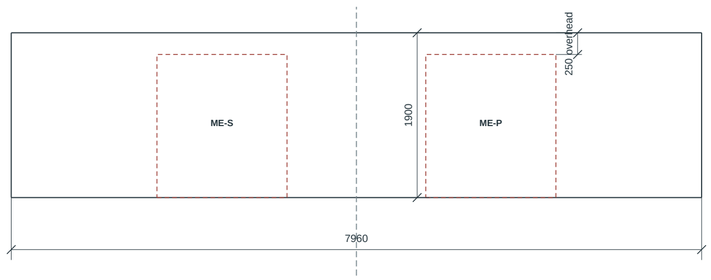


Car-deck nominal surface: Z 3000

Car-deck underside: Z 2800

Floor plates: Z 900

B-B TRANSVERSE SECTION AT X = 10700



Scale bar: 5 m at 1:50 (100% print)



SECTION QUALIFICATIONS

1. Red dashed rectangles: proposed 2700 x 1500 x 1650 mm combined machinery reservation, not the external outline of an approved engine.
2. Existing 1900 mm floor-to-deckhead is retained. Proposed machine top Z 2550 leaves only 250 mm nominal overhead space. A shoreside removal strategy is necessary unless a deeper hull or raised casing is engineered.
3. No deck beam, bracket, lagging, cable tray or structural scantling is dimensioned for fabrication. These remain release holds.
4. Both main engine centres are Y +/-1550 mm. Supplier 6AYM-WST reference width 1305 mm is wider than the current game block of 900 mm.
5. Envelope projection does not prove local collision freedom. A signed installation drawing and checked access model are required.

DATUMS AND PRINT CHECK

1. BL is Z 0; centreline is Y 0. Nominal laden waterline Z 1900; main deck Z 3000.
2. DXF profiles are XY drawing projections of X/Z physical values, and transverse sections are Y/Z projections. Three-dimensional source is separately supplied.
3. At 100% printing 1000 mm real distance occupies 20 mm on 1:50 sheets.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-102 | Machinery spaces - longitudinal and transverse sections

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 5

Corrected machinery reservation and maintenance strategy

PROPOSED layout reservations; retain present hull pending naval architecture and supplier review.

Red: yard reservation / removal cut-out. Teal: actual reference dimensions. Grey: desired service zones.



REFERENCE MACHINERY

1. Yanmar 6AYM-WST C operating reference (2020):485 kW at 1900 rpm, 6 cylinders, 20.379 L. Separate legacy H physical reference:engine2000 x 1305 x 1431 mm; dry 2365 kg.
2. Legacy YX181 gear reference:ratio3.03, dry 560 kg; published combined length2364 mm. Teal box uses combined length with engine-only width/height; full accessory extremes are unverified.
3. Fictional Johansson main remains 460 kW at 1350 rpm, 6 cylinders 160 x 200 mm, derived displacement 24.127 L. Its detailed installation drawing is unavailable.
4. Proposed combined reserve each: 2700 x 1500 x 1650 mm, 3.6 t estimated mass. Verification against an actual selected vendor assembly is a release hold.

MAINTENANCE AND REMOVAL HOLDS

1. Desired operational side access 1000 mm; non-operational side 800 mm; ends 600 mm. These are design targets, not OEM minima.
2. Each removable deck cut-out reserve 2900 x 1800 mm. Covers, coamings, fire integrity, deck strength, crane reach, rigging and lifting points require design and approval.
3. Only 250 mm nominal overhead clearance above the proposed envelope prevents approval of internal piston or head lifting.
4. Both gensets are relocated in the corrected source to auxiliary room X 15200 to16050. Control booth is retained. Model centre aisle1000 mm and side gap1035 mm are not maker clearances.
5. Grey targets overlap present allocations. This drawing records needed changes; it does not claim that the proposed arrangement is collision-free.

WHOLE-ASSEMBLY OVERHAUL CONCEPT

1. Vessel alongside, de-energised, fluids drained, shaft/cable/pipe interfaces disconnected under a planned procedure. Remove cover, lift to shore using engineered rigging, replace and align.
2. A crane WLL cannot be selected from mass alone. Account for rigging weight, radius, centre of gravity, dynamic allowance and verified lifting eyes.
3. References: Yanmar Norway legacy H physical datasheet and Yanmar C operating brochure (2020). Source identities differ; URLs and exact scopes are in reference_sources.csv.

Scale bar: 5 m at 1:50 (100% print)



Source e87c06b16734 | CAD units mm | dimensions govern
Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-103 | Corrected machinery reservation and maintenance strategy

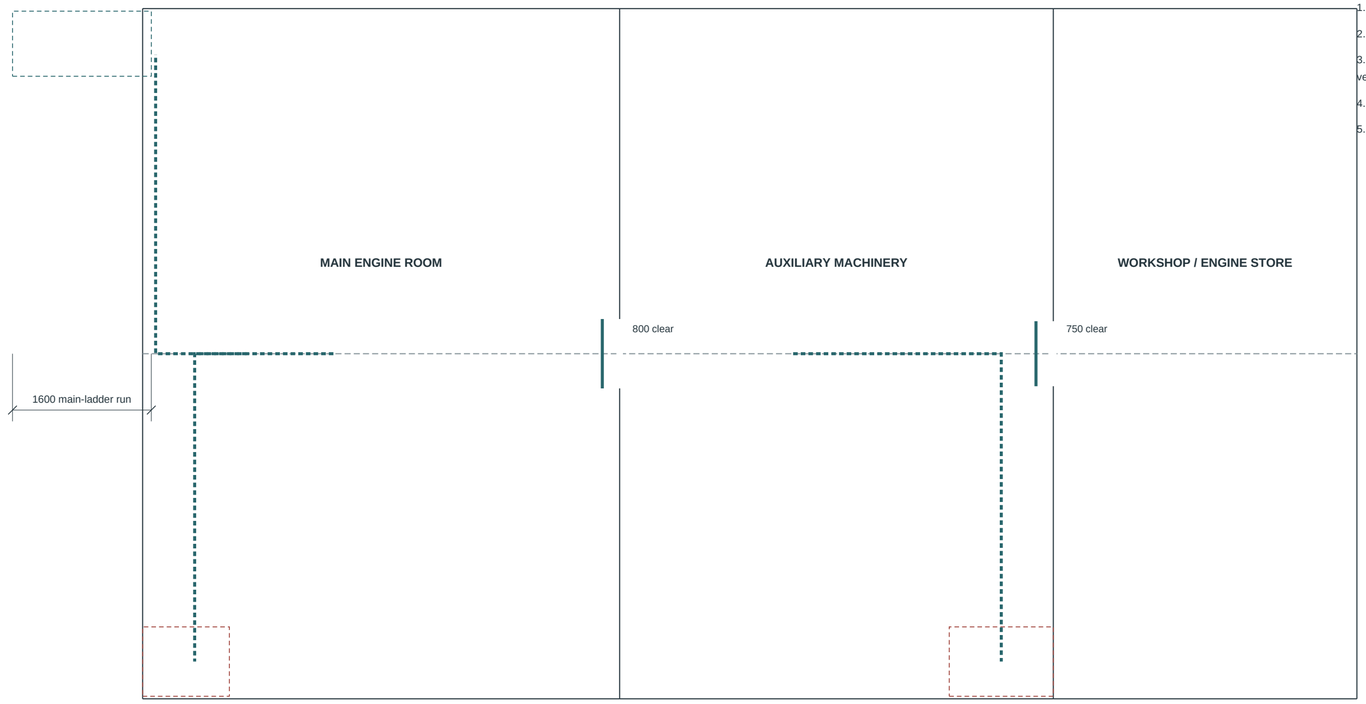
DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 6

Access, escape and compartment interfaces

Physical model-linked entrance, watertight doors and escape reservations; no evacuation certification.

Main ladder and two escape-trunk reservations (red); dotted paths are intended access connectivity.



ACCESS REVIEW

1. Main ladder source: X 7000 at upper end, X 8600 at foot; Y 3200 to3950. Source hatch X 7000 to9000, Y 3150 to4130.

2. ER / aux door at X 14000: 800 mm nominal width, 1900 mm height. Aux / workshop door at X 19000:750 mm nominal width, 1850 mm height.

3. Current aft-starboard escape ladder alone does not establish a continuous protected escape to the car deck. The red box is a proposed reservation; its complete trunk and hatch are not yet verified.

4. Auxiliary escape reserve at X 17800 to19000 on starboard. Direct route to casing must be kept clear from stored equipment.

5. No passenger escape-width, travel-distance, stair-angle or fire-boundary compliance is asserted. Applicable flag/class review remains open.

COMPARTMENT COORDINATION

1. Preserve two independent exits from attended machinery spaces. Keep watertight boundaries, quick-closing controls, fire isolation and escape continuity consistent across drawings and game geometry.

2. Red reservations must be coordinated with the existing control booth, passenger stair casing and pipe/cable penetrations. Never interpret a dotted intended path as an approved escape.

3. Minimum passage widths and doors are design dimensions. Regulatory adequacy requires the vessel-specific service basis and review.

Scale bar: 5 m at 1:50 (100% print)

Source e87c06b16734 | CAD units mm | dimensions govern
Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-104 | Access, escape and compartment interfaces

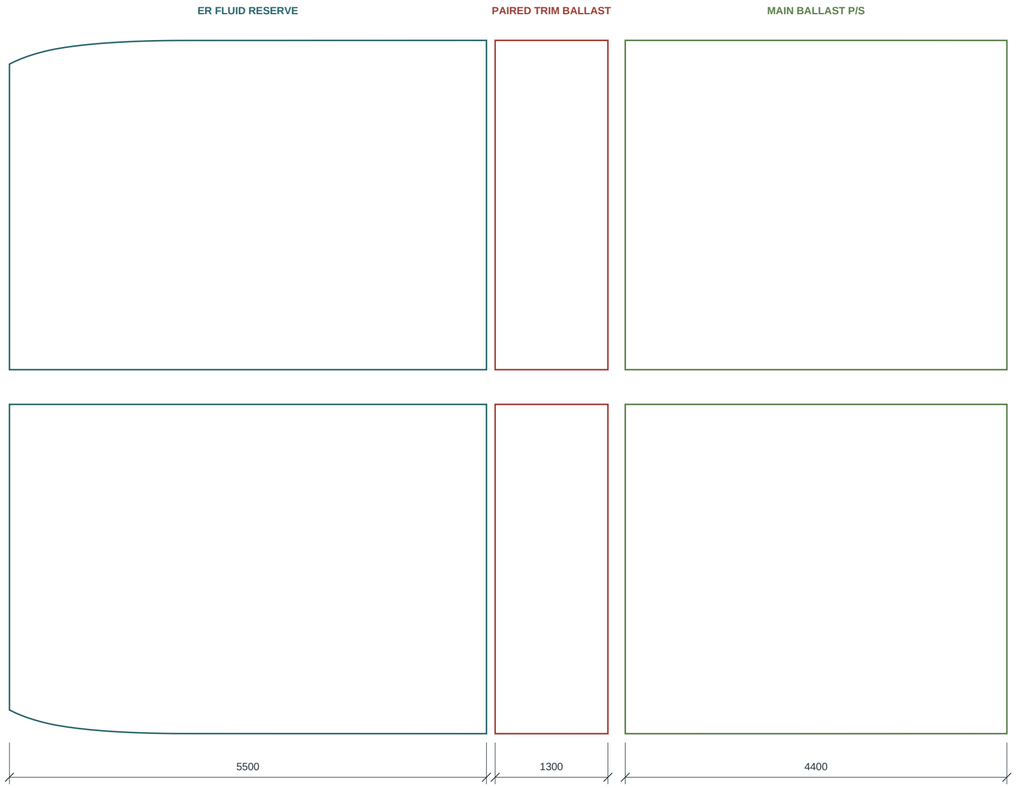
DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 7

Double-bottom tank allocation and conformal boundaries

Accepted planning reservations; 100 mm geometric clearance and10% volume deduction are not structural approval.

PLAN AT TANK UPPER LIMIT Z 680; gap400 along centreline



Allocation	X limits mm	Gross m3	Plan m3	Nominal m3
ER FLUID RESERVE	8500-14000	20.6140	18.5526	16.9
PAIRED TRIM BALLAST	14100-15400	5.3916	4.8524	4
MAIN BALLAST P/S	15600-20000	18.2487	16.4238	16

ALLOCATION AND CONSTRUCTION HOLDS

1. ER reserve accommodates16.9 m3 nominal fluids: 15 m3 fuel, 1 m3 bilge, 0.3 m3 sludge, 0.6 m3 lube oil. Individual partitions are not designed.
2. Trim reserve X 14100-15400: nominal 2 m3 each side. Main ballast X 15600-20000: nominal 8 m3 each side. Longitudinal zones are disjoint.
3. Outboard boundaries follow the loft minus100 mm horizontal reserve. Lower boundary is max(Z 100, hull keel+100). Upper limit Z 680.
4. 100 mm is geometric clearance, not plating thickness.10% volume deduction is a planning allowance, not calculated framing displacement or calibrated net capacity.
5. Partitions, scantlings, supports, welds, pressure tests, air/sounding pipes and free-surface calculations require engineering and approval.
6. Source ID aft-peak is retained for game compatibility; this adopted tank pair is under the auxiliary machinery room, not outside the stern hull.

TYPICAL TRIM TANK SECTION AT X 14750

X mm	Lower Z	Inner Y	OuterY Zlow	OuterY Z 390	OuterY Z 680
14100	100.0	200.0	3271.8	3832.0	3997.4
14500	100.0	200.0	3271.9	3832.1	3997.4
14750	100.0	200.0	3272.0	3832.1	3997.5
15000	100.0	200.0	3272.0	3832.2	3997.5
15400	100.0	200.0	3272.1	3832.2	3997.6

Scale bar: 5 m at 1:50 (100% print)

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

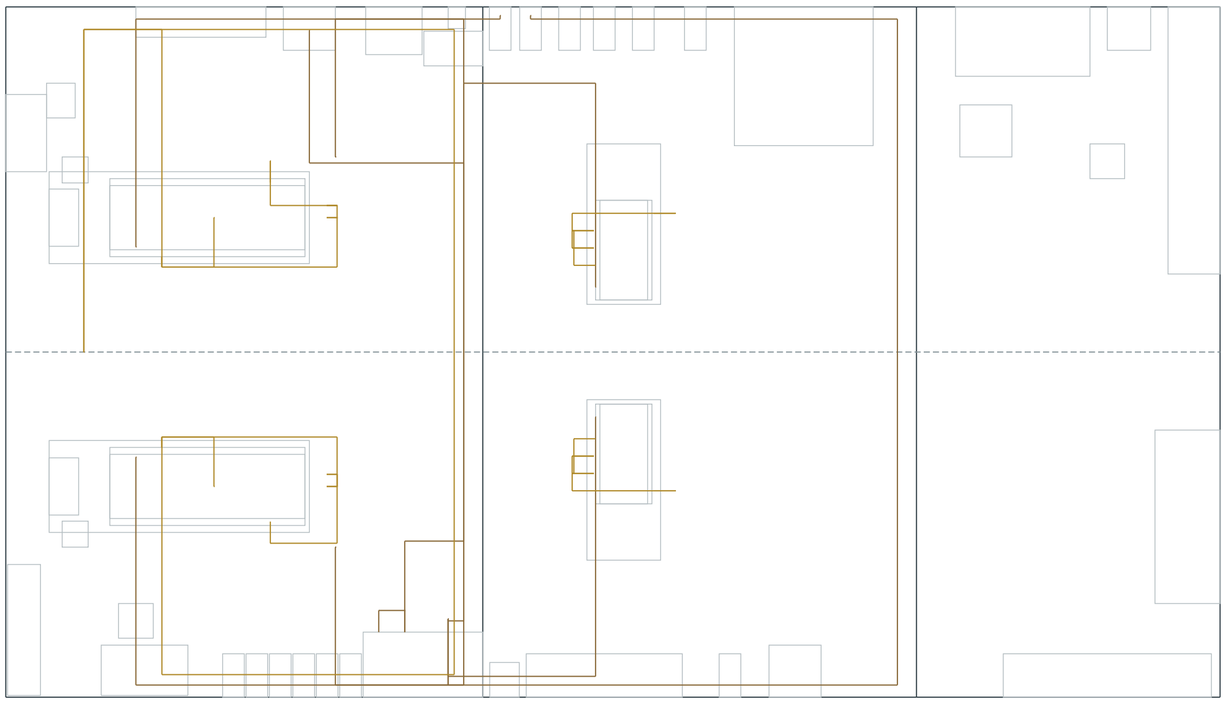
JS127-ER-108 | Double-bottom tank allocation and conformal boundaries

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 8

Physical pipe routes - fuel and lubricating oil

Plan projection of model pipe centre-lines; no pipe-class, support or clash approval.



Fuel oil

Lubricating oil

PHYSICAL ROUTING QUALIFICATIONS

1. 26 routed connections shown. Pipe coordinates and nominal diameters are exported in pipe_routes_mm.json; line/node schedule identifies missing physical connections.
2. Plan projection hides height crossings. Refer to route Z values and three-dimensional DXF/OBJ; equal plan locations do not prove pipe interference.
3. DN is nominal bore. Pipe wall, outside diameter, bends, flange patterns, valve face-to-face dimensions and insulation thickness remain unverified.
4. Penetrations require approved sleeves, seals, collars and watertight/fire integrity. A visible game flange is not a fabrication detail.
5. Routing currently excludes exhaust and hydraulics from the automatic physical lanes. Their connection schematics and release holds are supplied separately.
6. Only routes accepted by the current geometric screen are drawn here. Omitted connections remain unresolved; material, lagging, real fittings and support checks are additional release holds.

System	Model routes	Topology lines
fuel	10	22
lube	16	28

Scale bar: 5 m at 1:50 (100% print)

0 1 2 3 4 5

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

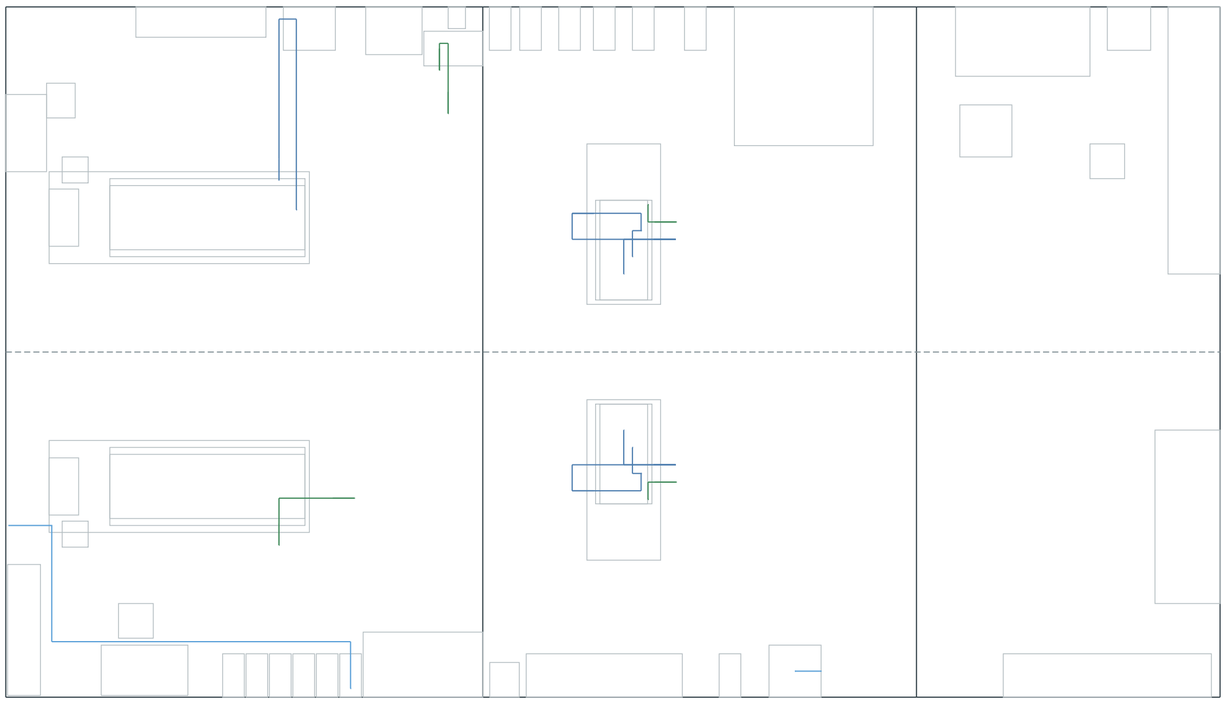
JS127-ER-105 | Physical pipe routes - fuel and lubricating oil

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 9

Physical pipe routes - cooling and water

Plan projection of model pipe centre-lines; no pipe-class, support or clash approval.



Sea-water cooling

Jacket water

Fresh water

PHYSICAL ROUTING QUALIFICATIONS

1. 15 routed connections shown. Pipe coordinates and nominal diameters are exported in pipe_routes_mm.json; line/node schedule identifies missing physical connections.
2. Plan projection hides height crossings. Refer to route Z values and three-dimensional DXF/OBJ; equal plan locations do not prove pipe interference.
3. DN is nominal bore. Pipe wall, outside diameter, bends, flange patterns, valve face-to-face dimensions and insulation thickness remain unverified.
4. Penetrations require approved sleeves, seals, collars and watertight/fire integrity. A visible game flange is not a fabrication detail.
5. Routing currently excludes exhaust and hydraulics from the automatic physical lanes. Their connection schematics and release holds are supplied separately.
6. Only routes accepted by the current geometric screen are drawn here. Omitted connections remain unresolved; material, lagging, real fittings and support checks are additional release holds.

System	Model routes	Topology lines
seawater	4	23
jacket	9	24
freshwater	2	15

Scale bar: 5 m at 1:50 (100% print)

0 1 2 3 4 5

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

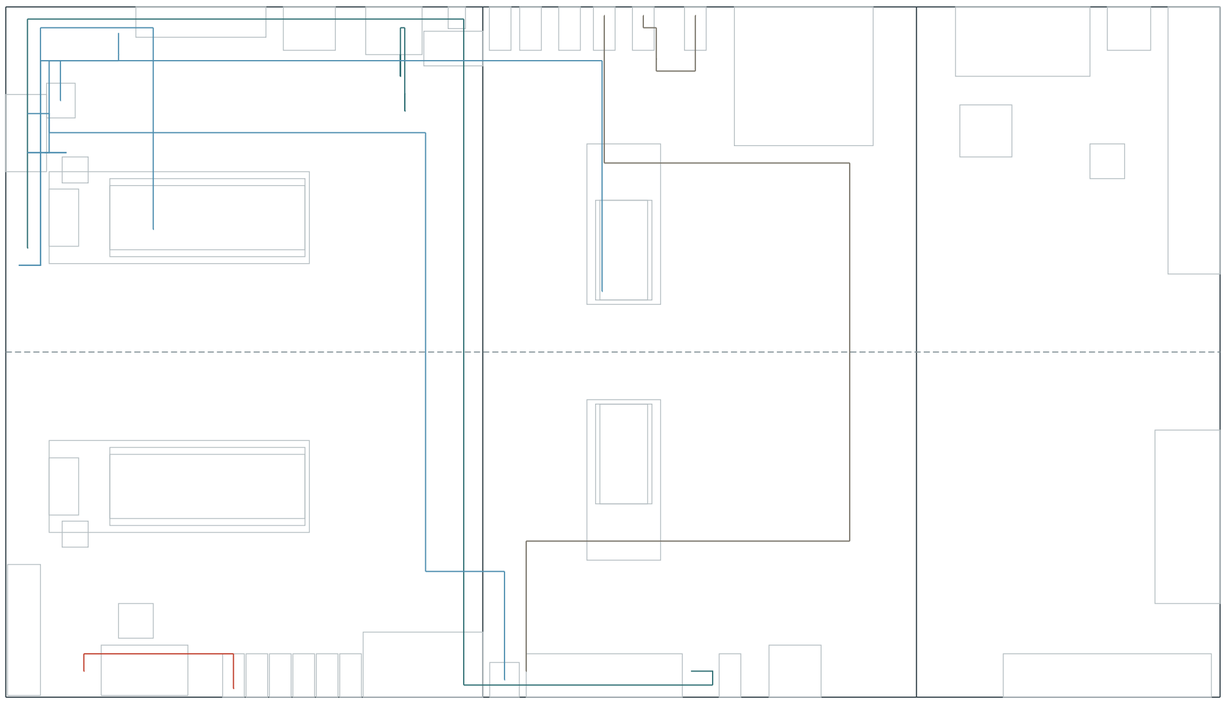
JS127-ER-106 | Physical pipe routes - cooling and water

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 10

Physical pipe routes - bilge, fire, air and sanitary

Plan projection of model pipe centre-lines; no pipe-class, support or clash approval.



- **Bilge and ballast**
- **Fire main and CO■**
- **Starting and service air**
- **Sanitary water and sewage**

PHYSICAL ROUTING QUALIFICATIONS

1. 14 routed connections shown. Pipe coordinates and nominal diameters are exported in pipe_routes_mm.json; line/node schedule identifies missing physical connections.
2. Plan projection hides height crossings. Refer to route Z values and three-dimensional DXF/OBJ; equal plan locations do not prove pipe interference.
3. DN is nominal bore. Pipe wall, outside diameter, bends, flange patterns, valve face-to-face dimensions and insulation thickness remain unverified.
4. Penetrations require approved sleeves, seals, collars and watertight/fire integrity. A visible game flange is not a fabrication detail.
5. Routing currently excludes exhaust and hydraulics from the automatic physical lanes. Their connection schematics and release holds are supplied separately.
6. Only routes accepted by the current geometric screen are drawn here. Omitted connections remain unresolved; material, lagging, real fittings and support checks are additional release holds.

System	Model routes	Topology lines
bilge	4	24
fire	1	22
air	6	17
sewage	3	10

Scale bar: 5 m at 1:50 (100% print)

0 1 2 3 4 5

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

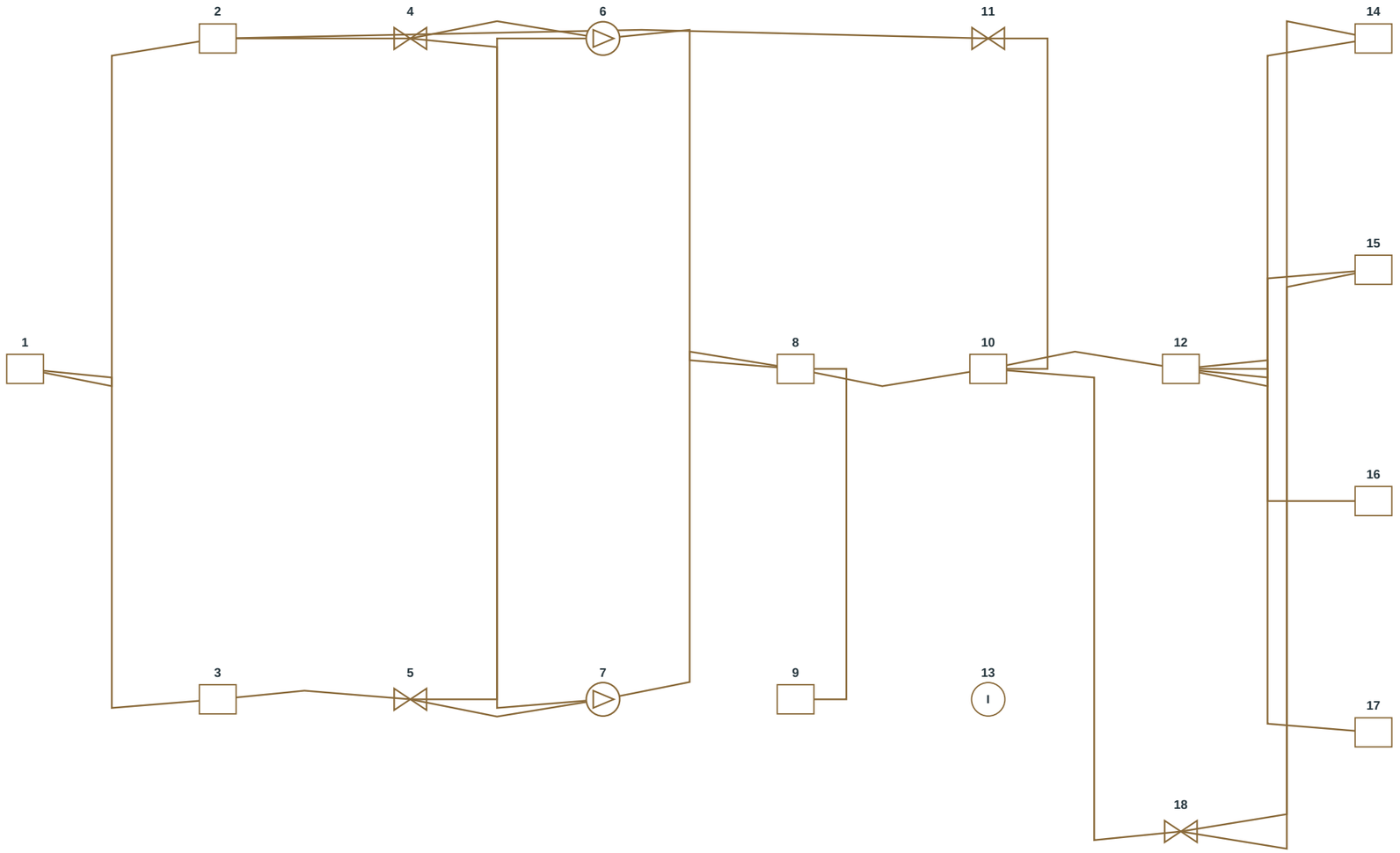
JS127-ER-107 | Physical pipe routes - bilge, fire, air and sanitary

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A2 landscape | SCALE 1:50 | SHEET 11

Fuel oil - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	bunker	10EGA10 AA501
2	storeP	10EGB10 BB001
3	storeS	10EGB10 BB002
4	vP	10EGC10 AA501
5	vS	10EGC10 AA502
6	tp1	10EGC10 AP001
7	tp2	10EGC10 AP002
8	purifier	10EGD10 AT001
9	sludge	10EGD10 BB002
10	service	10EGD20 BB001
11	overflow	10EGD20 AA401
12	qcv	10EGD20 AA201
13	lvl	10EGD20 CL001
14	fuel-mainP	11XJN10 AT001
15	fuel-mainS	12XJN10 AT001
16	fuel-gen1	13XJN10 AT001
17	fuel-gen2	14XJN10 AT001
18	return	10EGD20 AA503

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: A-heavy oil (marine distillate)

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-01 | Fuel oil - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 12

Fuel oil - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	bunker	10EGA10 AA501	Bunkering connection, port car deck	car-deck	MISSING
2	storeP	10EGB10 BB001	Fuel storage P · 7.5 m3	er-double-bottom	MODEL
3	storeS	10EGB10 BB002	Fuel storage S · 7.5 m3	er-double-bottom	MODEL
4	vP	10EGC10 AA501	Suction P	engine-room	MODEL
5	vS	10EGC10 AA502	Suction S	engine-room	MODEL
6	tp1	10EGC10 AP001	Fuel transfer pump No.1 · 1.5 kW · 2 m3/h	aux-room	MODEL
7	tp2	10EGC10 AP002	Fuel transfer pump No.2 · 1.5 kW · 2 m3/h	aux-room	MODEL
8	purifier	10EGD10 AT001	Fuel purifier JF-1 · 250 L/h	engine-room	MODEL
9	sludge	10EGD10 BB002	Sludge tank · 0.3 m3	er-double-bottom	MODEL
10	service	10EGD20 BB001	Service tank · 1.5 m3 · gauge glass · level alarm	engine-room	MODEL
11	overflow	10EGD20 AA401	Overflow to storage P	engine-room	MODEL
12	qcv	10EGD20 AA201	Quick-closing valve · wire pull from the CO■ room	engine-room	MODEL
13	lvl	10EGD20 CL001	Level alarm (FUEL LEVEL on the console)	engine-room	MISSING
14	fuel-mainP	11XJN10 AT001	Port main engine: filters, engine-driven feed pump, injection pumps	engine-room	MODEL
15	fuel-mainS	12XJN10 AT001	Starboard main engine: filters, engine-driven feed pump, injecti...	engine-room	MODEL
16	fuel-gen1	13XJN10 AT001	Generator No.1 engine: filters, engine-driven feed pump, injecti...	aux-room	MODEL
17	fuel-gen2	14XJN10 AT001	Generator No.2 engine: filters, engine-driven feed pump, injecti...	aux-room	MODEL
18	return	10EGD20 AA503	Return line to the service tank	engine-room	MODEL

SYSTEM NOTES

1. Transfer: No.1 and No.2 alternate weekly; either tank to either pump through the cross-suction.

2. Gravity from the service tank, 1.8 m over the engines' pumps; the quick-closing valve is shut from outside the engine room before CO■ is released.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-01-N1 | Fuel oil - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 13

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fuel oil - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
FUEL-001	bunker	storeP	50	MISSING	hose from the lorry; the filling line to each tank
FUEL-002	bunker	storeS	50	MISSING	
FUEL-003	storeP	vP	25	MODEL	
FUEL-004	storeS	vS	25	MODEL	
FUEL-005	vP	tp1	25	MODEL	
FUEL-006	vS	tp2	25	MODEL	
FUEL-007	vP	tp2	25	MISSING	cross-suction
FUEL-008	vS	tp1	25	MISSING	
FUEL-009	tp1	purifier	25	MISSING	
FUEL-010	tp2	purifier	25	MISSING	
FUEL-011	purifier	service	25	MODEL	
FUEL-012	purifier	sludge	20	MISSING	sludge discharge
FUEL-013	service	overflow	32	MODEL	
FUEL-014	overflow	storeP	32	MISSING	
FUEL-015	service	qcv	20	MISSING	
FUEL-016	qcv	fuel-mainP	20	MODEL	
FUEL-017	qcv	fuel-mainS	20	MODEL	
FUEL-018	qcv	fuel-gen1	15	MODEL	
FUEL-019	qcv	fuel-gen2	15	MODEL	
FUEL-020	fuel-mainP	return	15	MISSING	
FUEL-021	fuel-mainS	return	15	MISSING	
FUEL-022	return	service	15	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-01-L1 | Fuel oil - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

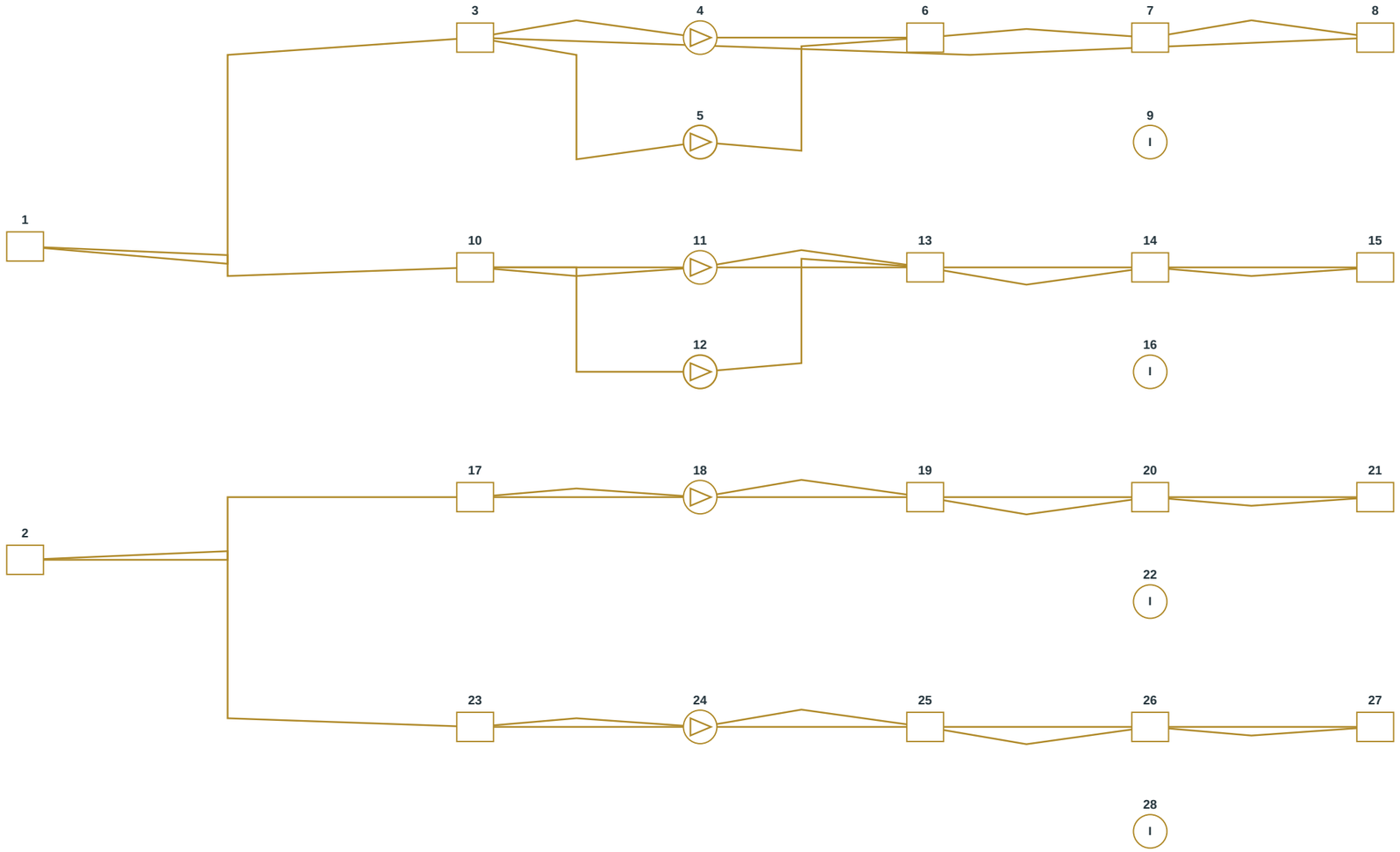
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 14

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Lubricating oil - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	lostore	10XJV10 BB001
2	drums	10XJV10 BB002
3	sump-mainP	11XJV10 BB003
4	lop-mainP	11XJV10 AP001
5	prime-mainP	11XJV10 AP002
6	lof-mainP	11XJV10 AT001
7	loc-mainP	11XJV10 AC001
8	bear-mainP	11XJA10 BR001
9	lop-mainP-p	11XJV10 CP001
10	sump-mainS	12XJV10 BB003
11	lop-mainS	12XJV10 AP001
12	prime-mainS	12XJV10 AP002
13	lof-mainS	12XJV10 AT001
14	loc-mainS	12XJV10 AC001
15	bear-mainS	12XJA10 BR001
16	lop-mainS-p	12XJV10 CP001
17	sump-gen1	13XJV10 BB003
18	lop-gen1	13XJV10 AP001
19	lof-gen1	13XJV10 AT001
20	loc-gen1	13XJV10 AC001
21	bear-gen1	13XJA10 BR001
22	lop-gen1-p	13XJV10 CP001
23	sump-gen2	14XJV10 BB003
24	lop-gen2	14XJV10 AP001
25	lof-gen2	14XJV10 AT001
26	loc-gen2	14XJV10 AC001
27	bear-gen2	14XJA10 BR001
28	lop-gen2-p	14XJV10 CP001

Medium: SAE 40 engine oil

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-02 | Lubricating oil - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 15

Lubricating oil - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	lostore	10XJV10 BB001	Lubricating-oil storage · 0.6 m3 · hand pump	er-double-bottom	MODEL
2	drums	10XJV10 BB002	Two 200 L drums on their rack, drip tray and taps	engine-store	MISSING
3	sump-mainP	11XJV10 BB003	Port main engine sump	engine-room	MODEL
4	lop-mainP	11XJV10 AP001	Engine-driven lube-oil pump	engine-room	MODEL
5	prime-mainP	11XJV10 AP002	Priming pump · 0.75 kW (before the first start)	engine-room	MODEL
6	lof-mainP	11XJV10 AT001	Duplex filter · changed by the hours	engine-room	MODEL
7	loc-mainP	11XJV10 AC001	Lube-oil cooler (sea water)	engine-room	MODEL
8	bear-mainP	11XJA10 BR001	Bearings, camshaft, turbocharger	engine-room	MODEL
9	lop-mainP-p	11XJV10 CP001	LO pressure (LO PRESS on the console; shutdown)	engine-room	MISSING
10	sump-mainS	12XJV10 BB003	Starboard main engine sump	engine-room	MODEL
11	lop-mainS	12XJV10 AP001	Engine-driven lube-oil pump	engine-room	MODEL
12	prime-mainS	12XJV10 AP002	Priming pump · 0.75 kW (before the first start)	engine-room	MODEL
13	lof-mainS	12XJV10 AT001	Duplex filter · changed by the hours	engine-room	MODEL
14	loc-mainS	12XJV10 AC001	Lube-oil cooler (sea water)	engine-room	MODEL
15	bear-mainS	12XJA10 BR001	Bearings, camshaft, turbocharger	engine-room	MODEL
16	lop-mainS-p	12XJV10 CP001	LO pressure (LO PRESS on the console; shutdown)	engine-room	MISSING
17	sump-gen1	13XJV10 BB003	Generator No.1 engine sump	aux-room	MODEL
18	lop-gen1	13XJV10 AP001	Engine-driven lube-oil pump	aux-room	MODEL
19	lof-gen1	13XJV10 AT001	Duplex filter · changed by the hours	aux-room	MODEL
20	loc-gen1	13XJV10 AC001	Lube-oil cooler (sea water)	aux-room	MODEL

SYSTEM NOTES

1. Engine-driven pumps run with the engine, electricity or not; the priming pumps fill the galleries before the first start of the day (electrical load lube-priming).

2. The gensets have no priming pump: a hand pump on each.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-02-N1 | Lubricating oil - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 16

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Lubricating oil - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
21	bear-gen1	13XJA10 BR001	Bearings, camshaft, turbocharger	aux-room	MODEL
22	lop-gen1-p	13XJV10 CP001	LO pressure (LO PRESS on the console; shutdown)	aux-room	MISSING
23	sump-gen2	14XJV10 BB003	Generator No.2 engine sump	aux-room	MODEL
24	lop-gen2	14XJV10 AP001	Engine-driven lube-oil pump	aux-room	MODEL
25	lof-gen2	14XJV10 AT001	Duplex filter · changed by the hours	aux-room	MODEL
26	loc-gen2	14XJV10 AC001	Lube-oil cooler (sea water)	aux-room	MODEL
27	bear-gen2	14XJA10 BR001	Bearings, camshaft, turbocharger	aux-room	MODEL
28	lop-gen2-p	14XJV10 CP001	LO pressure (LO PRESS on the console; shutdown)	aux-room	MISSING

SYSTEM NOTES

1. Engine-driven pumps run with the engine, electricity or not; the priming pumps fill the galleries before the first start of the day (electrical load lube-priming).

2. The gensets have no priming pump: a hand pump on each.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-02-N2 | Lubricating oil - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 17

Lubricating oil - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
LUBE-001	lostore	sump-mainP	20	MODEL	top-up by the hand pump
LUBE-002	lostore	sump-mainS	20	MODEL	
LUBE-003	drums	sump-gen1	signal	MISSING	a can at a time
LUBE-004	drums	sump-gen2	signal	MISSING	
LUBE-005	sump-mainP	lop-mainP	40	MODEL	
LUBE-006	sump-mainP	prime-mainP	25	MISSING	
LUBE-007	prime-mainP	lof-mainP	25	MISSING	
LUBE-008	lop-mainP	lof-mainP	40	MODEL	
LUBE-009	lof-mainP	loc-mainP	40	MODEL	
LUBE-010	loc-mainP	bear-mainP	40	MISSING	
LUBE-011	bear-mainP	sump-mainP	50	MODEL	drain to the sump
LUBE-012	sump-mainS	lop-mainS	40	MODEL	
LUBE-013	sump-mainS	prime-mainS	25	MISSING	
LUBE-014	prime-mainS	lof-mainS	25	MISSING	
LUBE-015	lop-mainS	lof-mainS	40	MODEL	
LUBE-016	lof-mainS	loc-mainS	40	MODEL	
LUBE-017	loc-mainS	bear-mainS	40	MISSING	
LUBE-018	bear-mainS	sump-mainS	50	MODEL	drain to the sump
LUBE-019	sump-gen1	lop-gen1	40	MODEL	
LUBE-020	lop-gen1	lof-gen1	40	MODEL	
LUBE-021	lof-gen1	loc-gen1	40	MODEL	
LUBE-022	loc-gen1	bear-gen1	40	MISSING	
LUBE-023	bear-gen1	sump-gen1	50	MISSING	drain to the sump
LUBE-024	sump-gen2	lop-gen2	40	MODEL	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-02-L1 | Lubricating oil - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 18

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Lubricating oil - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
LUBE-025	lop-gen2	lof-gen2	40	MODEL	
LUBE-026	lof-gen2	loc-gen2	40	MODEL	
LUBE-027	loc-gen2	bear-gen2	40	MISSING	
LUBE-028	bear-gen2	sump-gen2	50	MISSING	drain to the sump

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-02-L2 | Lubricating oil - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

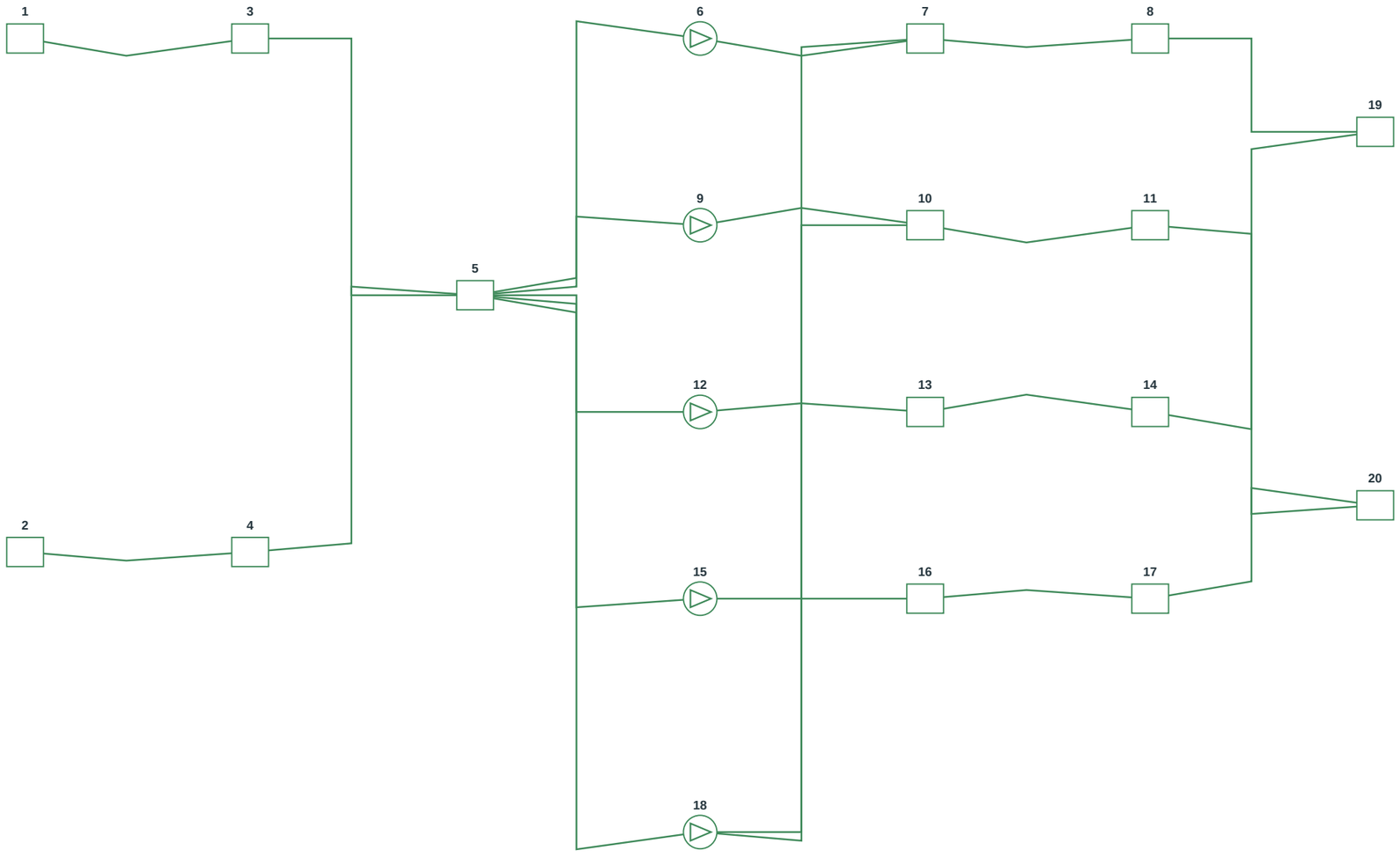
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 19

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Sea-water cooling - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	chestP	10PAB10 AA501
2	chestS	10PAB10 AA502
3	strP	10PAB10 AT001
4	strS	10PAB10 AT002
5	cross	10PAB10 AA503
6	swp-mainP	11PAB20 AP001
7	cac-mainP	11XJH10 AC001
8	jwc-mainP	11XJG10 AC001
9	swp-mainS	12PAB20 AP001
10	cac-mainS	12XJH10 AC001
11	jwc-mainS	12XJG10 AC001
12	swp-gen1	13PAB20 AP001
13	cac-gen1	13XJH10 AC001
14	jwc-gen1	13XJG10 AC001
15	swp-gen2	14PAB20 AP001
16	cac-gen2	14XJH10 AC001
17	jwc-gen2	14XJG10 AC001
18	gs	10GAA10 AP001
19	obP	10PAB30 AA504
20	obS	10PAB30 AA505

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Sea water

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-03 | Sea-water cooling - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 20

Sea-water cooling - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	chestP	10PAB10 AA501	Sea chest P (high, under the plates)	engine-room	MODEL
2	chestS	10PAB10 AA502	Sea chest S (low)	engine-room	MODEL
3	strP	10PAB10 AT001	Strainer P · basket Mondays	engine-room	MODEL
4	strS	10PAB10 AT002	Strainer S	engine-room	MODEL
5	cross	10PAB10 AA503	Crossover: either chest to every pump	engine-room	MODEL
6	swp-mainP	11PAB20 AP001	Port main engine: engine-driven sea-water pump	engine-room	MODEL
7	cac-mainP	11XJH10 AC001	Charge-air cooler	engine-room	MODEL
8	jwc-mainP	11XJG10 AC001	Jacket-water cooler	engine-room	MODEL
9	swp-mainS	12PAB20 AP001	Starboard main engine: engine-driven sea-water pump	engine-room	MODEL
10	cac-mainS	12XJH10 AC001	Charge-air cooler	engine-room	MODEL
11	jwc-mainS	12XJG10 AC001	Jacket-water cooler	engine-room	MODEL
12	swp-gen1	13PAB20 AP001	Generator No.1 engine: engine-driven sea-water pump	aux-room	MODEL
13	cac-gen1	13XJH10 AC001	Charge-air cooler	aux-room	MODEL
14	jwc-gen1	13XJG10 AC001	Jacket-water cooler	aux-room	MODEL
15	swp-gen2	14PAB20 AP001	Generator No.2 engine: engine-driven sea-water pump	aux-room	MODEL
16	cac-gen2	14XJH10 AC001	Charge-air cooler	aux-room	MODEL
17	jwc-gen2	14XJG10 AC001	Jacket-water cooler	aux-room	MODEL
18	gs	10GAA10 AP001	General-service pump JP-32 · 11 kW · standby cooling through the...	engine-room	MODEL
19	obP	10PAB30 AA504	Overboard P · non-return	engine-room	MODEL
20	obS	10PAB30 AA505	Overboard S · non-return	engine-room	MODEL

SYSTEM NOTES

1. Each engine cools itself as long as it runs; the general-service pump takes over either main engine's cooling through the crossover if its own pump fails.
2. The lube-oil coolers (8.2) sit in the same sea-water run after the charge-air coolers.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-03-N1 | Sea-water cooling - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 21

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Sea-water cooling - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
SEAWATER-001	chestP	strP	65	MISSING	
SEAWATER-002	chestS	strS	65	MISSING	
SEAWATER-003	strP	cross	65	MODEL	
SEAWATER-004	strS	cross	65	MISSING	
SEAWATER-005	cross	swp-mainP	50	MISSING	
SEAWATER-006	swp-mainP	cac-mainP	50	MISSING	
SEAWATER-007	cac-mainP	jwc-mainP	50	MISSING	
SEAWATER-008	jwc-mainP	obP	50	MISSING	
SEAWATER-009	cross	swp-mainS	50	MISSING	
SEAWATER-010	swp-mainS	cac-mainS	50	MISSING	
SEAWATER-011	cac-mainS	jwc-mainS	50	MODEL	
SEAWATER-012	jwc-mainS	obS	50	MISSING	
SEAWATER-013	cross	swp-gen1	32	MISSING	
SEAWATER-014	swp-gen1	cac-gen1	32	MODEL	
SEAWATER-015	cac-gen1	jwc-gen1	32	MISSING	
SEAWATER-016	jwc-gen1	obP	32	MISSING	
SEAWATER-017	cross	swp-gen2	32	MISSING	
SEAWATER-018	swp-gen2	cac-gen2	32	MODEL	
SEAWATER-019	cac-gen2	jwc-gen2	32	MISSING	
SEAWATER-020	jwc-gen2	obS	32	MISSING	
SEAWATER-021	cross	gs	65	MISSING	
SEAWATER-022	gs	cac-mainP	50	MISSING	standby to either main engine
SEAWATER-023	gs	cac-mainS	50	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-03-L1 | Sea-water cooling - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

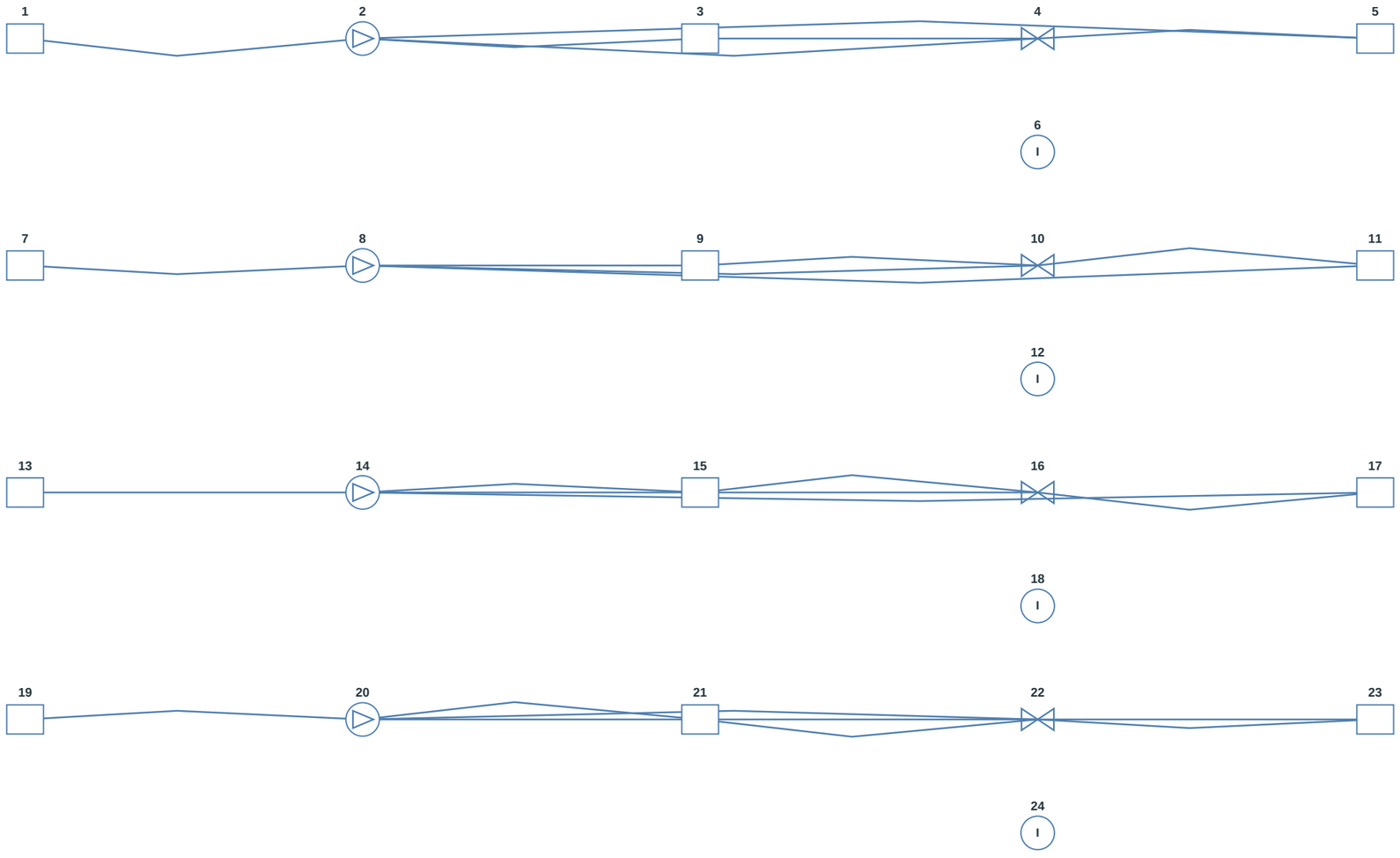
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 22

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Jacket water - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	exp-mainP	11XJG10 BB001
2	jwp-mainP	11XJG10 AP001
3	blk-mainP	11XJA10 BR002
4	thermo-mainP	11XJG10 AA601
5	jwc2-mainP	11XJG10 AC001
6	jwt-mainP	11XJG10 CT001
7	exp-mainS	12XJG10 BB001
8	jwp-mainS	12XJG10 AP001
9	blk-mainS	12XJA10 BR002
10	thermo-mainS	12XJG10 AA601
11	jwc2-mainS	12XJG10 AC001
12	jwt-mainS	12XJG10 CT001
13	exp-gen1	13XJG10 BB001
14	jwp-gen1	13XJG10 AP001
15	blk-gen1	13XJA10 BR002
16	thermo-gen1	13XJG10 AA601
17	jwc2-gen1	13XJG10 AC001
18	jwt-gen1	13XJG10 CT001
19	exp-gen2	14XJG10 BB001
20	jwp-gen2	14XJG10 AP001
21	blk-gen2	14XJA10 BR002
22	thermo-gen2	14XJG10 AA601
23	jwc2-gen2	14XJG10 AC001
24	jwt-gen2	14XJG10 CT001

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Fresh water with inhibitor

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-04 | Jacket water - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 23

Jacket water - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	exp-mainP	11XJG10 BB001	Port main engine: expansion tank with sight glass and low-level ...	engine-room	MODEL
2	jwp-mainP	11XJG10 AP001	Engine-driven jacket-water pump	engine-room	MODEL
3	blk-mainP	11XJA10 BR002	Cylinder block, heads, turbocharger	engine-room	MODEL
4	thermo-mainP	11XJG10 AA601	Thermostat · 82 deg C	engine-room	MODEL
5	jwc2-mainP	11XJG10 AC001	Jacket-water cooler (sea water: 8.3)	engine-room	MODEL
6	jwt-mainP	11XJG10 CT001	Water temperature (WATER on the console; shutdown)	engine-room	MISSING
7	exp-mainS	12XJG10 BB001	Starboard main engine: expansion tank with sight glass and low-l...	engine-room	MODEL
8	jwp-mainS	12XJG10 AP001	Engine-driven jacket-water pump	engine-room	MODEL
9	blk-mainS	12XJA10 BR002	Cylinder block, heads, turbocharger	engine-room	MODEL
10	thermo-mainS	12XJG10 AA601	Thermostat · 82 deg C	engine-room	MODEL
11	jwc2-mainS	12XJG10 AC001	Jacket-water cooler (sea water: 8.3)	engine-room	MODEL
12	jwt-mainS	12XJG10 CT001	Water temperature (WATER on the console; shutdown)	engine-room	MISSING
13	exp-gen1	13XJG10 BB001	Generator No.1 engine: expansion tank with sight glass and low-l...	aux-room	MODEL
14	jwp-gen1	13XJG10 AP001	Engine-driven jacket-water pump	aux-room	MODEL
15	blk-gen1	13XJA10 BR002	Cylinder block, heads, turbocharger	aux-room	MODEL
16	thermo-gen1	13XJG10 AA601	Thermostat · 82 deg C	aux-room	MODEL
17	jwc2-gen1	13XJG10 AC001	Jacket-water cooler (sea water: 8.3)	aux-room	MODEL
18	jwt-gen1	13XJG10 CT001	Water temperature (WATER on the console; shutdown)	aux-room	MISSING
19	exp-gen2	14XJG10 BB001	Generator No.2 engine: expansion tank with sight glass and low-l...	aux-room	MODEL
20	jwp-gen2	14XJG10 AP001	Engine-driven jacket-water pump	aux-room	MODEL

SYSTEM NOTES

1. No electric pump: the circuit lives and dies with its engine. Filled from the fresh-water system (8.8) through the expansion tank.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-04-N1 | Jacket water - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 24

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Jacket water - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
21	blk-gen2	14XJA10 BR002	Cylinder block, heads, turbocharger	aux-room	MODEL
22	thermo-gen2	14XJG10 AA601	Thermostat · 82 deg C	aux-room	MODEL
23	jwc2-gen2	14XJG10 AC001	Jacket-water cooler (sea water: 8.3)	aux-room	MODEL
24	jwt-gen2	14XJG10 CT001	Water temperature (WATER on the console; shutdown)	aux-room	MISSING

SYSTEM NOTES

1. No electric pump: the circuit lives and dies with its engine. Filled from the fresh-water system (8.8) through the expansion tank.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-04-N2 | Jacket water - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 25

Jacket water - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
JACKET-001	exp-mainP	jwp-mainP	25	MISSING	static head
JACKET-002	jwp-mainP	blk-mainP	50	MISSING	
JACKET-003	blk-mainP	thermo-mainP	50	MISSING	
JACKET-004	thermo-mainP	jwc2-mainP	50	MODEL	
JACKET-005	jwc2-mainP	jwp-mainP	50	MISSING	
JACKET-006	thermo-mainP	jwp-mainP	40	MISSING	bypass while cold
JACKET-007	exp-mainS	jwp-mainS	25	MISSING	static head
JACKET-008	jwp-mainS	blk-mainS	50	MISSING	
JACKET-009	blk-mainS	thermo-mainS	50	MISSING	
JACKET-010	thermo-mainS	jwc2-mainS	50	MISSING	
JACKET-011	jwc2-mainS	jwp-mainS	50	MISSING	
JACKET-012	thermo-mainS	jwp-mainS	40	MISSING	bypass while cold
JACKET-013	exp-gen1	jwp-gen1	25	MODEL	static head
JACKET-014	jwp-gen1	blk-gen1	50	MISSING	
JACKET-015	blk-gen1	thermo-gen1	50	MODEL	
JACKET-016	thermo-gen1	jwc2-gen1	50	MODEL	
JACKET-017	jwc2-gen1	jwp-gen1	50	MODEL	
JACKET-018	thermo-gen1	jwp-gen1	40	MISSING	bypass while cold
JACKET-019	exp-gen2	jwp-gen2	25	MODEL	static head
JACKET-020	jwp-gen2	blk-gen2	50	MISSING	
JACKET-021	blk-gen2	thermo-gen2	50	MODEL	
JACKET-022	thermo-gen2	jwc2-gen2	50	MODEL	
JACKET-023	jwc2-gen2	jwp-gen2	50	MODEL	
JACKET-024	thermo-gen2	jwp-gen2	40	MISSING	bypass while cold

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-04-L1 | Jacket water - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

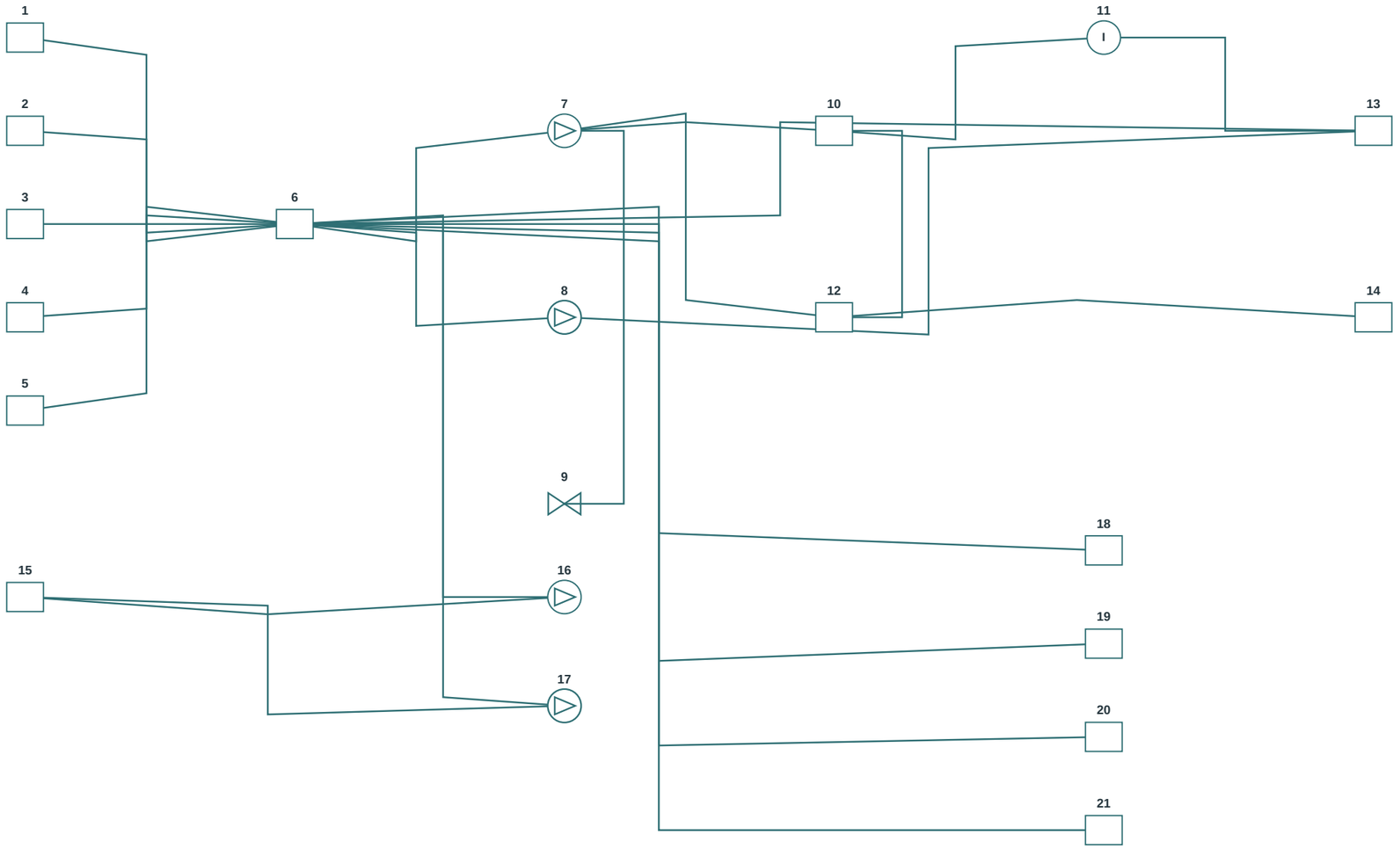
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 26

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Bilge and ballast - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	wellERP	10GMA10 AA501
2	wellERS	10GMA10 AA502
3	wellAux	10GMA10 AA503
4	wellShaft	10GMA10 AA504
5	wellSG	10GMA10 AA505
6	manifold	10GMA10 AA506
7	bp	10GMA20 AP001
8	fbp	10SGA10 AP002
9	emerg	10GMA10 AA507
10	ows	10GNA10 AT001
11	ppm	10GNA10 CQ001
12	hold	10GMA20 BB001
13	ob	10GNA10 AA504
14	shoreB	10GMA20 AA508
15	chest	10PAB10 AA503
16	balp	10GHA10 AP001
17	gs2	10GAA10 AP001
18	balP	10GHA20 BB001
19	balS	10GHA20 BB002
20	aftpk	10GHA20 BB003
21	forepk	10GHA20 BB004

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Bilge water · sea water

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-05 | Bilge and ballast - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 27

Bilge and ballast - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	wellERP	10GMA10 AA501	Bilge well ER aft P	engine-room	MODEL
2	wellERS	10GMA10 AA502	Bilge well ER aft S	engine-room	MODEL
3	wellAux	10GMA10 AA503	Bilge well aux room	aux-room	MODEL
4	wellShaft	10GMA10 AA504	Shaft space	shaft-space	MODEL
5	wellSG	10GMA10 AA505	Steering-gear room	steering-gear	MODEL
6	manifold	10GMA10 AA506	Bilge and ballast manifold · six valves	aux-room	MODEL
7	bp	10GMA20 AP001	Bilge pump JP-25 · 3.7 kW · 25 m3/h	engine-room	MODEL
8	fbp	10SGA10 AP002	Fire and bilge pump No.2 (8.6) · 11 kW	engine-room	MODEL
9	emerg	10GMA10 AA507	Emergency bilge suction on the port main engine's sea-water pump	engine-room	MODEL
10	ows	10GNA10 AT001	Oily-water separator JO-15 · 0.25 m3/h · 15 ppm	engine-room	MODEL
11	ppm	10GNA10 CQ001	15 ppm monitor · stops the overboard	engine-room	MODEL
12	hold	10GMA20 BB001	Bilge holding tank · 1 m3	er-double-bottom	MODEL
13	ob	10GNA10 AA504	Overboard · non-return · 15 ppm interlock	engine-room	MODEL
14	shoreB	10GMA20 AA508	Shore connection (bilge, sludge) · car deck	car-deck	MODEL
15	chest	10PAB10 AA503	Sea chest (8.3 crossover)	engine-room	MODEL
16	balp	10GHA10 AP001	Ballast pump JP-32 · 3.7 kW	aux-room	MODEL
17	gs2	10GAA10 AP001	General-service pump (standby, 8.3)	engine-room	MODEL
18	balP	10GHA20 BB001	Ballast P · 8 m3	fwd-double-bottom	MODEL
19	balS	10GHA20 BB002	Ballast S · 8 m3	fwd-double-bottom	MODEL
20	aftpk	10GHA20 BB003	Aft peak · 4 m3	aft-peak	MODEL

SYSTEM NOTES

1. The bilge main is 65 mm (25 + 1.68·√(L(B+D))); branches 50 and 40. Two pumps on it, and the emergency suction on a main engine's pump.

2. Bilge water goes overboard only through the separator and its 15 ppm monitor; in port it goes to the holding tank and ashore.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-05-N1 | Bilge and ballast - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 28

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Bilge and ballast - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
21	forepk	10GHA20 BB004	Fore peak · 6 m3	fore-peak	MODEL

SYSTEM NOTES

1. The bilge main is 65 mm (25 + 1.68·√(L(B+D))); branches 50 and 40. Two pumps on it, and the emergency suction on a main engine's pump.

2. Bilge water goes overboard only through the separator and its 15 ppm monitor; in port it goes to the holding tank and ashore.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-05-N2 | Bilge and ballast - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 29

Bilge and ballast - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
BILGE-001	wellERP	manifold	50	MODEL	
BILGE-002	wellERS	manifold	50	MISSING	
BILGE-003	wellAux	manifold	50	MISSING	
BILGE-004	wellShaft	manifold	40	MISSING	
BILGE-005	wellSG	manifold	40	MISSING	
BILGE-006	manifold	bp	65	MISSING	
BILGE-007	manifold	fbp	65	MISSING	
BILGE-008	emerg	bp	65	MISSING	direct suction, no separator: only to save the ship
BILGE-009	bp	ows	25	MISSING	
BILGE-010	bp	hold	65	MISSING	
BILGE-011	fbp	ob	65	MISSING	fire-and-bilge pump straight overboard: emergencies only
BILGE-012	ows	ppm	25	MODEL	
BILGE-013	ppm	ob	25	MISSING	
BILGE-014	ows	hold	25	MODEL	oil and dirty water back to the tank
BILGE-015	hold	shoreB	50	MODEL	
BILGE-016	chest	balp	65	MISSING	
BILGE-017	chest	gs2	65	MISSING	
BILGE-018	balp	manifold	65	MISSING	
BILGE-019	gs2	manifold	65	MISSING	
BILGE-020	manifold	balP	65	MISSING	
BILGE-021	manifold	balS	65	MISSING	
BILGE-022	manifold	aftpk	50	MISSING	
BILGE-023	manifold	forepk	50	MISSING	
BILGE-024	manifold	ob	65	MISSING	ballast discharge

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-05-L1 | Bilge and ballast - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

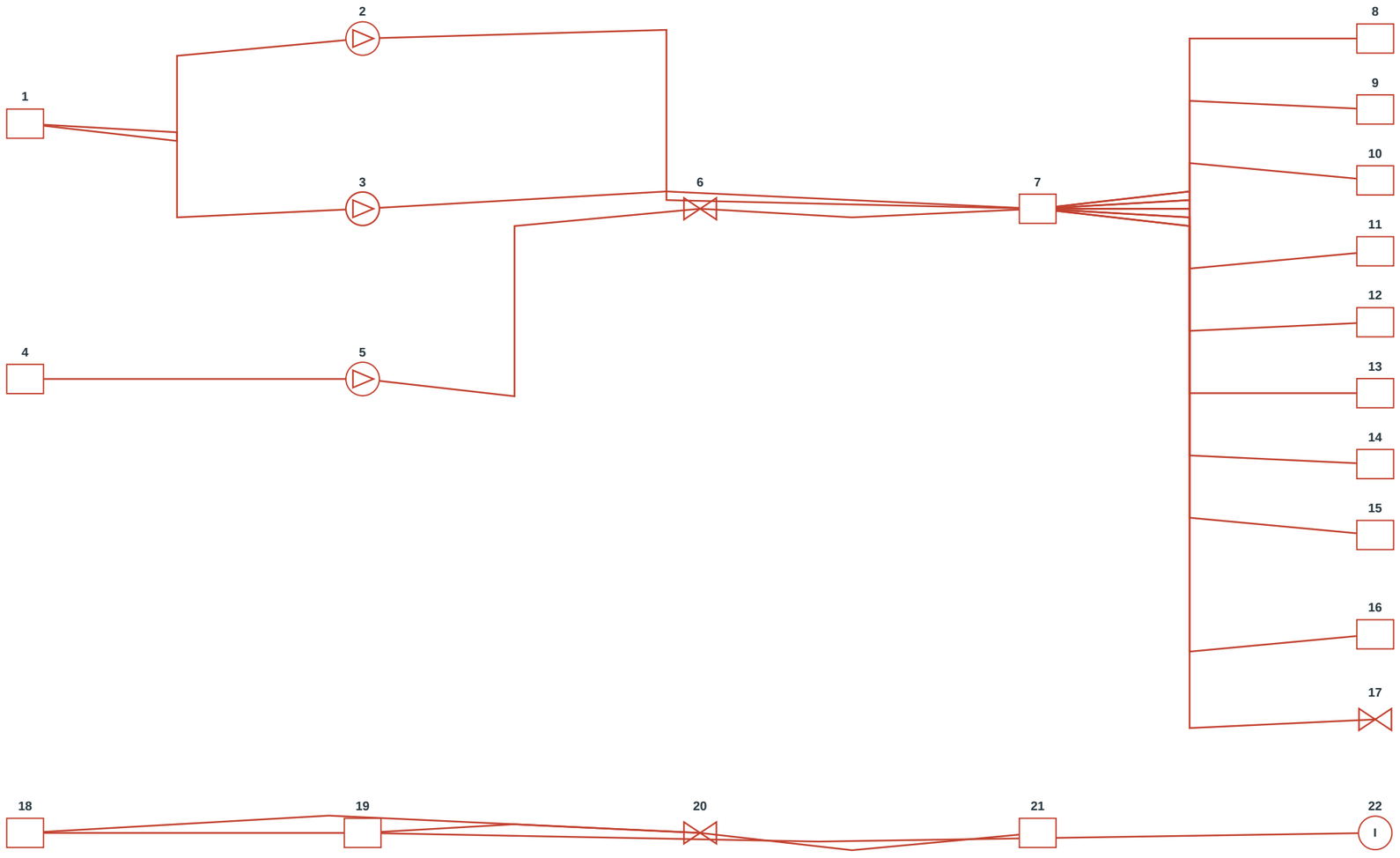
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 30

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fire main and CO■ - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	fchest	10PAB10 AA503
2	fp1	10SGA10 AP001
3	fp2	10SGA10 AP002
4	efchest	10SGA20 AA501
5	efp	10SGA20 AP001
6	iso	10SGA30 AA502
7	main	10SGA30 BR001
8	hER	10SGA30 AA511
9	hCar1	10SGA30 AA512
10	hCar2	10SGA30 AA513
11	hCar3	10SGA30 AA514
12	hCar4	10SGA30 AA515
13	hPassP	10SGA30 AA516
14	hPassS	10SGA30 AA517
15	hBridge	10SGA30 AA518
16	isc	10SGA30 AA519
17	wash	10SGA30 AA520
18	co2b	10SGJ10 BB001
19	co2rel	10SGJ10 CH001
20	co2v	10SGJ10 AA201
21	co2n	10SGJ10 BR001
22	co2al	10SGJ10 CH002

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Sea water · CO■

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-06 | Fire main and CO■ - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 31

Fire main and CO■ - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	fchest	10PAB10 AA503	Sea chests (8.3 crossover)	engine-room	MODEL
2	fp1	10SGA10 AP001	Fire and GS pump No.1 JP-40 · 11 kW · 40 m3/h at 0.5 MPa	engine-room	MODEL
3	fp2	10SGA10 AP002	Fire and bilge pump No.2 JP-40 · 11 kW	engine-room	MODEL
4	efchest	10SGA20 AA501	Own sea suction, steering-gear room	steering-gear	MISSING
5	efp	10SGA20 AP001	Emergency fire pump · 5.5 kW · 20 m3/h · emergency switchboard	steering-gear	MISSING
6	iso	10SGA30 AA502	Isolating valve: the engine room cut off, the emergency pump fee...	steering-gear	MISSING
7	main	10SGA30 BR001	Fire main · 65 mm · 0.5 MPa	car-deck	MODEL
8	hER	10SGA30 AA511	Hydrant engine room · 15 m hose · 12 mm nozzle	engine-room	MODEL
9	hCar1	10SGA30 AA512	Hydrant car deck P aft · 15 m hose · 12 mm nozzle	car-deck	MODEL
10	hCar2	10SGA30 AA513	Hydrant car deck S aft · 15 m hose · 12 mm nozzle	car-deck	MODEL
11	hCar3	10SGA30 AA514	Hydrant car deck P fwd · 15 m hose · 12 mm nozzle	car-deck	MODEL
12	hCar4	10SGA30 AA515	Hydrant car deck S fwd · 15 m hose · 12 mm nozzle	car-deck	MODEL
13	hPassP	10SGA30 AA516	Hydrant passenger deck P · 15 m hose · 12 mm nozzle	walkway-p	MODEL
14	hPassS	10SGA30 AA517	Hydrant passenger deck S · 15 m hose · 12 mm nozzle	walkway-s	MODEL
15	hBridge	10SGA30 AA518	Hydrant bridge deck · 15 m hose · 12 mm nozzle	wing-p	MODEL
16	isc	10SGA30 AA519	International shore connection · port car deck	car-deck	MODEL
17	wash	10SGA30 AA520	Deck wash and anchor wash	bow-house-p	MODEL
18	co2b	10SGJ10 BB001	4 x 45 kg CO■	co2-room	MISSING
19	co2rel	10SGJ10 CH001	Release cabinet · key · alarm first	co2-room	MISSING
20	co2v	10SGJ10 AA201	Main valve · opened by the cabinet	co2-room	MODEL

SYSTEM NOTES

1. Three pumps: two in the engine room, one outside it on the emergency board with its own sea suction, so a fire in the engine room never takes the fire main with it.

2. CO■: Provisional inventory screen only: 180 kg carried against an assumed 190 m3 protected volume. Actual fire boundaries, ducts, protected volumes, nozzle distribution and release interlocks require engineered verification.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-06-N1 | Fire main and CO■ - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 32

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fire main and CO■ - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
21	co2n	10SGJ10 BR001	Six nozzles under the engine-room deckhead	engine-room	MODEL
22	co2al	10SGJ10 CH002	CO■ alarm and fan stop before release	engine-room	MISSING

SYSTEM NOTES

1. Three pumps: two in the engine room, one outside it on the emergency board with its own sea suction, so a fire in the engine room never takes the fire main with it.

2. CO■: Provisional inventory screen only: 180 kg carried against an assumed 190 m3 protected volume. Actual fire boundaries, ducts, protected volumes, nozzle distribution and release interlocks require engineered verification.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-06-N2 | Fire main and CO■ - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 33

Fire main and CO■ - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
FIRE-001	fchest	fp1	65	MISSING	
FIRE-002	fchest	fp2	65	MISSING	
FIRE-003	efchest	efp	65	MISSING	
FIRE-004	fp1	main	65	MODEL	
FIRE-005	fp2	main	65	MISSING	
FIRE-006	efp	iso	65	MISSING	
FIRE-007	iso	main	65	MISSING	
FIRE-008	main	hER	50	MISSING	
FIRE-009	main	hCar1	50	MISSING	
FIRE-010	main	hCar2	50	MISSING	
FIRE-011	main	hCar3	50	MISSING	
FIRE-012	main	hCar4	50	MISSING	
FIRE-013	main	hPassP	50	MISSING	
FIRE-014	main	hPassS	50	MISSING	
FIRE-015	main	hBridge	50	MISSING	
FIRE-016	main	isc	65	MISSING	
FIRE-017	main	wash	50	MISSING	
FIRE-018	co2b	co2rel	signal	MISSING	wire
FIRE-019	co2rel	co2v	25	MISSING	pilot
FIRE-020	co2b	co2v	40	MISSING	
FIRE-021	co2v	co2n	40	MISSING	
FIRE-022	co2rel	co2al	signal	MISSING	electric: the alarm sounds and the fans stop

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-06-L1 | Fire main and CO■ - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

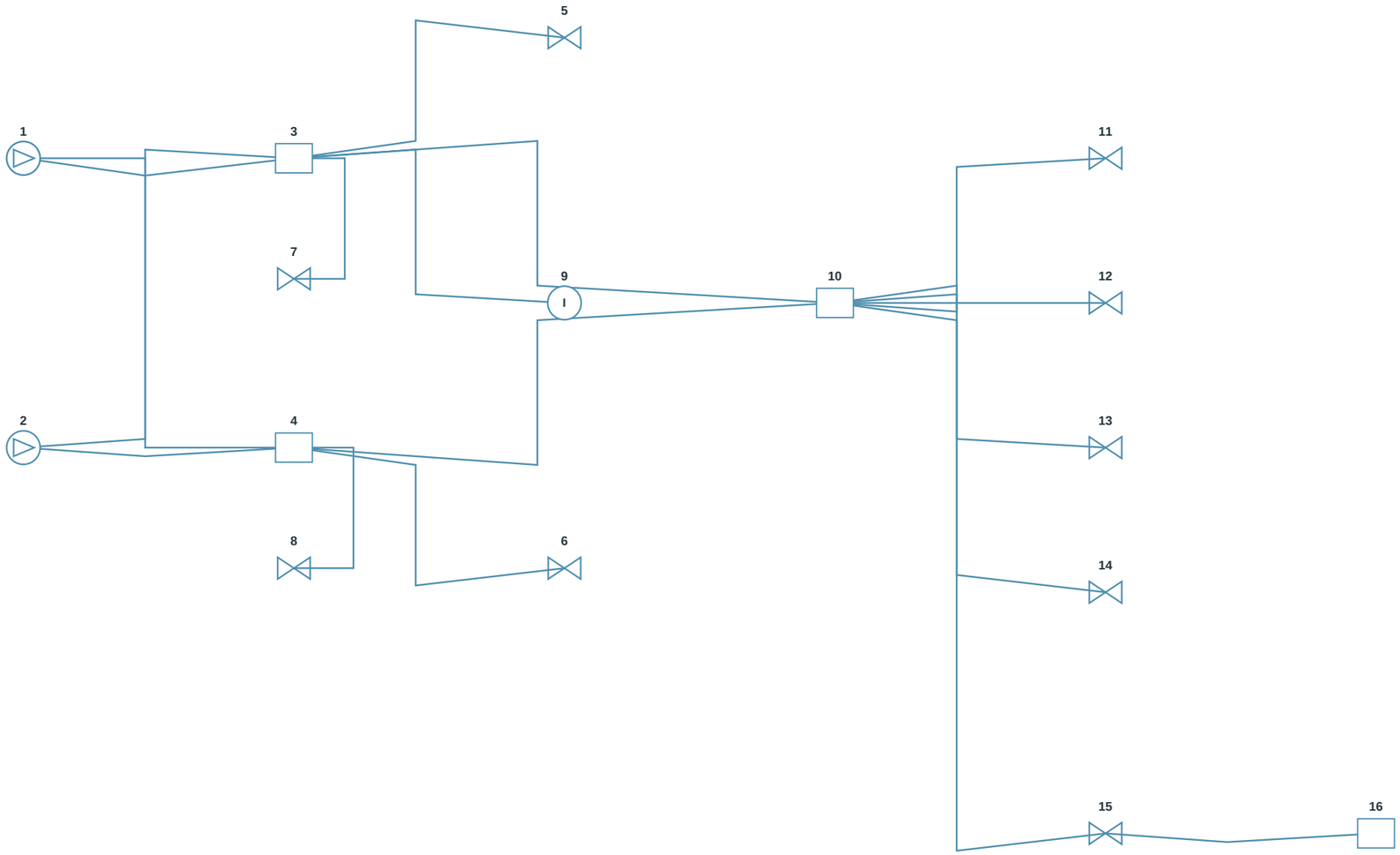
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 34

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Starting and service air - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	c1	10SCA10 AN001
2	c2	10SCA10 AN002
3	b1	10SCA20 BB001
4	b2	10SCA20 BB002
5	sv1	10SCA20 AA901
6	sv2	10SCA20 AA902
7	dr1	10SCA20 AA401
8	dr2	10SCA20 AA402
9	pr	10SCA20 CP301
10	dist	10SCB10 BR001
11	start-mainP	11XJJ10 AA201
12	start-mainS	12XJJ10 AA201
13	start-gen1	13XJJ10 AA201
14	start-gen2	14XJJ10 AA201
15	red	10SCB20 AA001
16	svc	10SCB20 BR001

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Compressed air · 30 bar / 7 bar

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-07 | Starting and service air - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 35

Starting and service air - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	c1	10SCA10 AN001	Starting-air compressor No.1 JC-30 · 3.7 kW	engine-room	MODEL
2	c2	10SCA10 AN002	Starting-air compressor No.2 JC-30 · 3.7 kW	aux-room	MODEL
3	b1	10SCA20 BB001	Air bottle No.1 · 150 L · 30 bar	engine-room	MODEL
4	b2	10SCA20 BB002	Air bottle No.2 · 150 L · 30 bar	engine-room	MODEL
5	sv1	10SCA20 AA901	Safety valve 33 bar	engine-room	MODEL
6	sv2	10SCA20 AA902	Safety valve 33 bar	engine-room	MODEL
7	dr1	10SCA20 AA401	Drain (blown down each watch)	engine-room	MODEL
8	dr2	10SCA20 AA402	Drain	engine-room	MODEL
9	pr	10SCA20 CP301	Pressure switch: start at 25, stop at 30 bar	engine-room	MODEL
10	dist	10SCB10 BR001	Starting-air main · 30 bar	engine-room	MODEL
11	start-mainP	11XJJ10 AA201	Port main engine: air-start valve and distributor	engine-room	MODEL
12	start-mainS	12XJJ10 AA201	Starboard main engine: air-start valve and distributor	engine-room	MODEL
13	start-gen1	13XJJ10 AA201	Generator No.1 engine: air-start valve and distributor	aux-room	MODEL
14	start-gen2	14XJJ10 AA201	Generator No.2 engine: air-start valve and distributor	aux-room	MODEL
15	red	10SCB20 AA001	Reducing station 30 -> 7 bar · service air	engine-room	MODEL
16	svc	10SCB20 BR001	Service air: the workshop, the whistle, the horn	workshop	MODEL

SYSTEM NOTES

1. Two compressors and two bottles: the rule asks for two means of starting; each bottle starts a main engine six times from cold.

2. Compressor No.1 and No.2 alternate weekly on the pressure switch.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-07-N1 | Starting and service air - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 36

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Starting and service air - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
AIR-001	c1	b1	20	MODEL	
AIR-002	c2	b2	20	MODEL	
AIR-003	c1	b2	20	MISSING	either compressor to either bottle
AIR-004	c2	b1	20	MISSING	
AIR-005	b1	sv1	15	MISSING	
AIR-006	b2	sv2	15	MISSING	
AIR-007	b1	dr1	10	MODEL	
AIR-008	b2	dr2	10	MISSING	
AIR-009	b1	pr	10	MISSING	
AIR-010	b1	dist	25	MISSING	
AIR-011	b2	dist	25	MISSING	
AIR-012	dist	start-mainP	25	MODEL	
AIR-013	dist	start-mainS	25	MISSING	
AIR-014	dist	start-gen1	15	MODEL	
AIR-015	dist	start-gen2	15	MISSING	
AIR-016	dist	red	15	MODEL	
AIR-017	red	svc	15	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-07-L1 | Starting and service air - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

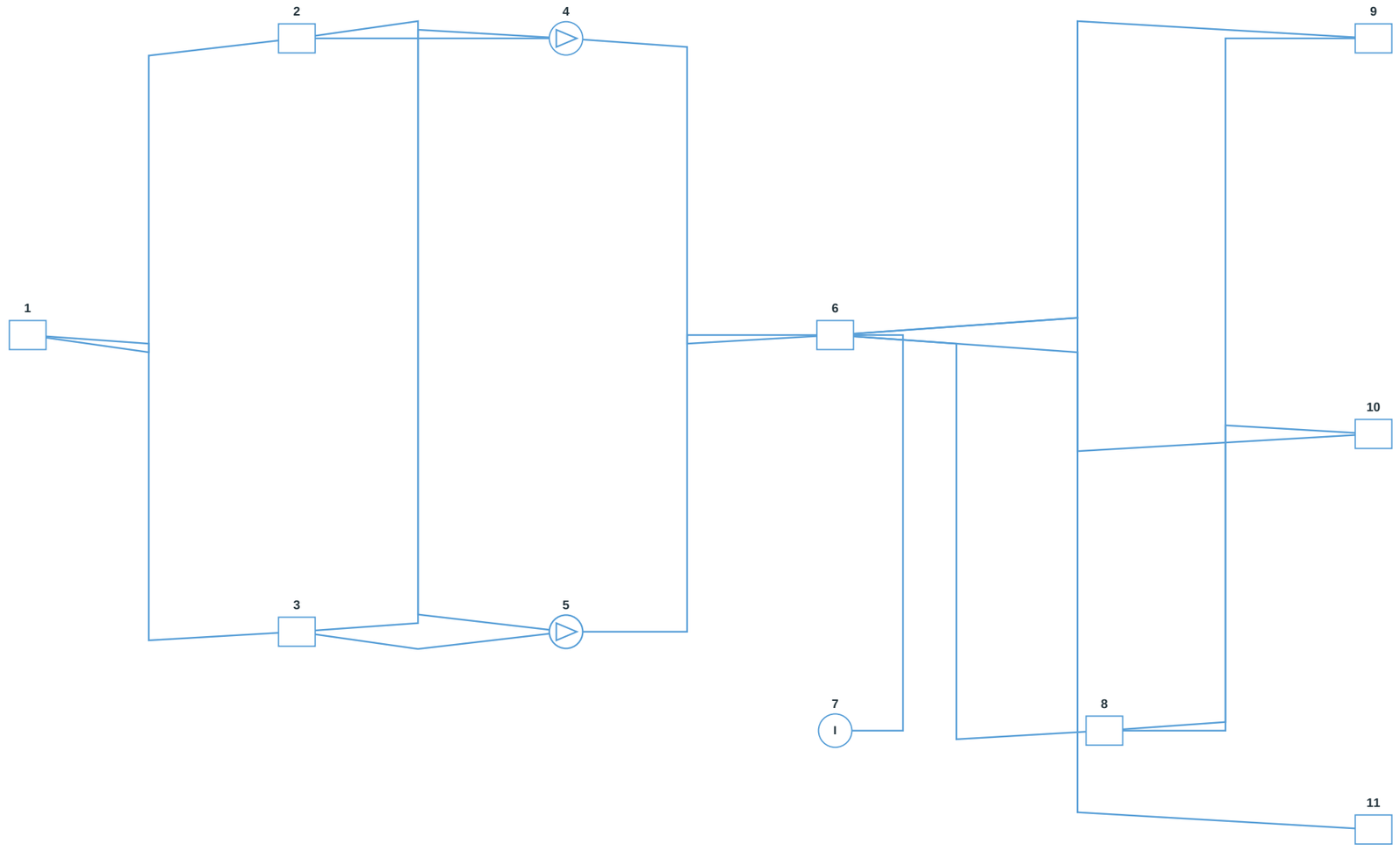
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 37

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fresh water - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	fill	10GKA10 AA501
2	fwP	10GKA10 BB001
3	fwS	10GKA10 BB002
4	hp1	10GKB10 AP001
5	hp2	10GKB10 AP002
6	ptank	10GKB10 BB001
7	psw	10GKB10 CP301
8	heater	10GKB20 AC001
9	galley	10GKB20 BR001
10	toilets	10GKB20 BR002
11	expfill	10GKB20 BR003

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Drinking water

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-08 | Fresh water - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 38

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fresh water - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	fill	10GKA10 AA501	Filling connection · Mr Chinen's water lorry, every fifth morning	car-deck	MISSING
2	fwP	10GKA10 BB001	Fresh-water tank P · 6 m3	fresh-water	MODEL
3	fwS	10GKA10 BB002	Fresh-water tank S · 6 m3	fresh-water	MODEL
4	hp1	10GKB10 AP001	Hydrophore pump No.1 · 0.75 kW	engine-room	MODEL
5	hp2	10GKB10 AP002	Hydrophore pump No.2 · 0.75 kW	aux-room	MODEL
6	ptank	10GKB10 BB001	Pressure tank · 300 L · 0.25-0.35 MPa	engine-room	MODEL
7	psw	10GKB10 CP301	Pressure switch (cut in, cut out)	engine-room	MODEL
8	heater	10GKB20 AC001	Water heater · 100 L · 3 kW	galley	MODEL
9	galley	10GKB20 BR001	Galley · kiosk hot water	galley	MODEL
10	toilets	10GKB20 BR002	Toilets' basins · crew WC and shower	toilets	MODEL
11	expfill	10GKB20 BR003	Expansion tanks (8.4) · the ER tap	engine-room	MODEL

SYSTEM NOTES

1. Two pumps on one pressure tank, alternating; 1.5 m3 a day for the taps.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-08-N1 | Fresh water - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 39

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Fresh water - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
FRESHWATER-001	fill	fwP	40	MISSING	
FRESHWATER-002	fill	fwS	40	MISSING	
FRESHWATER-003	fwP	hp1	32	MISSING	
FRESHWATER-004	fwS	hp1	32	MODEL	
FRESHWATER-005	fwP	hp2	32	MISSING	
FRESHWATER-006	fwS	hp2	32	MISSING	
FRESHWATER-007	hp1	ptank	32	MISSING	
FRESHWATER-008	hp2	ptank	32	MISSING	
FRESHWATER-009	ptank	psw	10	MODEL	
FRESHWATER-010	ptank	galley	20	MISSING	
FRESHWATER-011	ptank	toilets	20	MISSING	
FRESHWATER-012	ptank	heater	15	MISSING	
FRESHWATER-013	heater	galley	15	MISSING	hot
FRESHWATER-014	heater	toilets	15	MISSING	hot
FRESHWATER-015	ptank	expfill	15	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-08-L1 | Fresh water - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

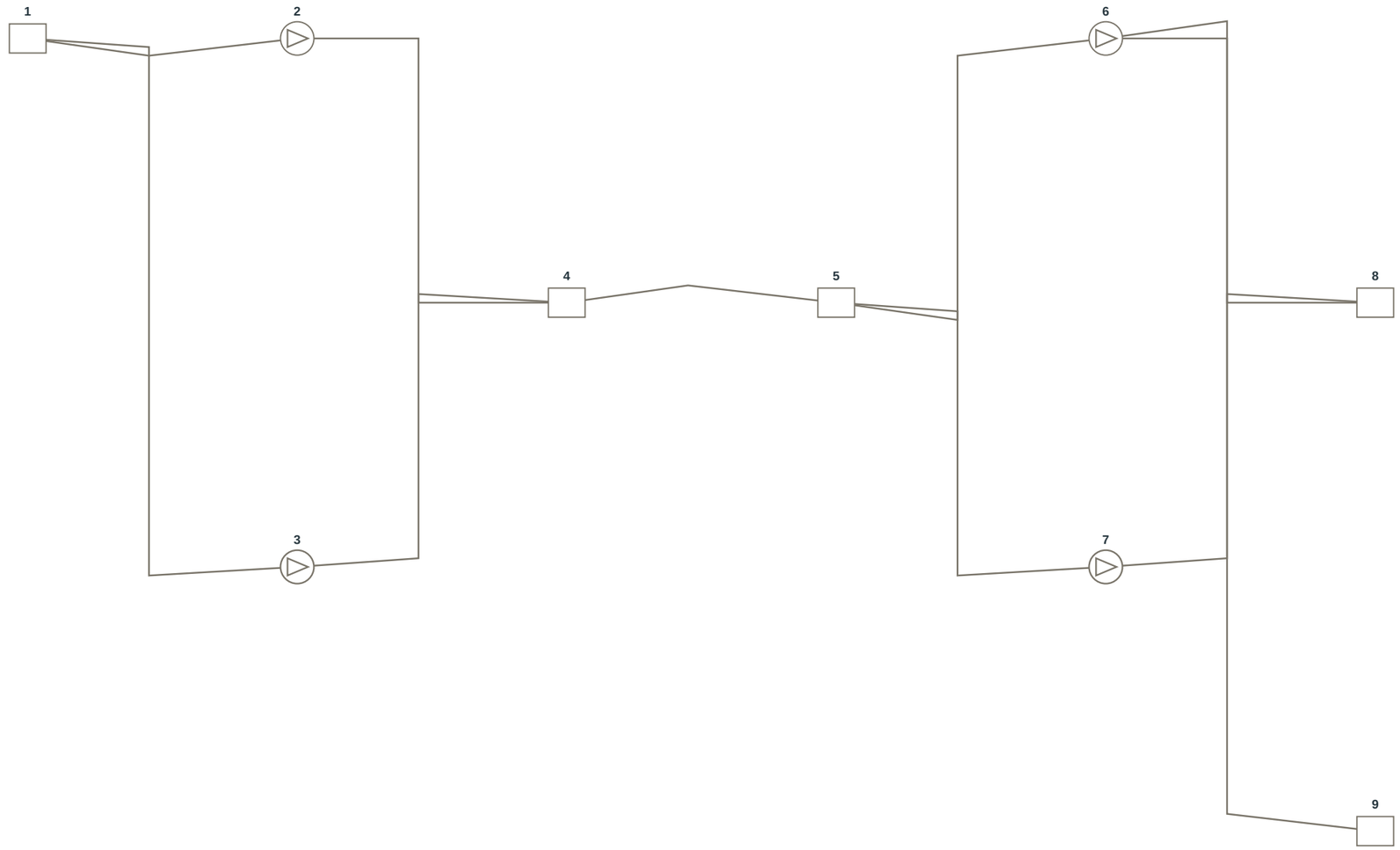
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 40

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Sanitary water and sewage - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	schest	10PAB10 AA503
2	sp1	10GHB10 AP001
3	sp2	10GHB10 AP002
4	flush	10GHB10 BR001
5	htank	10GQA10 BB001
6	dp1	10GRA10 AP001
7	dp2	10GRA10 AP002
8	deck	10GRA10 AA501
9	ob2	10GRA10 AA502

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Sea water · sewage

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-09 | Sanitary water and sewage - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 41

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Sanitary water and sewage - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	schest	10PAB10 AA503	Sea chest (8.3 crossover)	engine-room	MODEL
2	sp1	10GHB10 AP001	Sanitary pump No.1 JP-15 · 1.5 kW	engine-room	MODEL
3	sp2	10GHB10 AP002	Sanitary pump No.2 JP-15 · 1.5 kW	aux-room	MODEL
4	flush	10GHB10 BR001	Flushing: three WCs, two urinals, the crew WC	toilets	MODEL
5	htank	10GQA10 BB001	Sewage holding tank · 4 m3 · sight glass · vent up the casing	aux-room	MODEL
6	dp1	10GRA10 AP001	Sewage discharge pump No.1 JP-15 · 1.5 kW	aux-room	MODEL
7	dp2	10GRA10 AP002	Sewage discharge pump No.2 JP-15 · 1.5 kW	aux-room	MODEL
8	deck	10GRA10 AA501	Deck connection to the quay's sewage truck · after the last boat	car-deck	MODEL
9	ob2	10GRA10 AA502	Overboard · blanked and sealed (never inside the strait)	aux-room	MODEL

SYSTEM NOTES

1. Nothing goes into the strait: the tank is pumped to the truck at Minato; the overboard is sealed with the inspector's seal.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-09-N1 | Sanitary water and sewage - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 42

Sanitary water and sewage - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
SEWAGE-001	schest	sp1	32	MISSING	
SEWAGE-002	schest	sp2	32	MISSING	
SEWAGE-003	sp1	flush	32	MISSING	
SEWAGE-004	sp2	flush	32	MODEL	
SEWAGE-005	flush	htank	100	MISSING	gravity
SEWAGE-006	htank	dp1	50	MISSING	
SEWAGE-007	htank	dp2	50	MISSING	
SEWAGE-008	dp1	deck	50	MODEL	
SEWAGE-009	dp2	deck	50	MODEL	
SEWAGE-010	dp1	ob2	50	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-09-L1 | Sanitary water and sewage - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

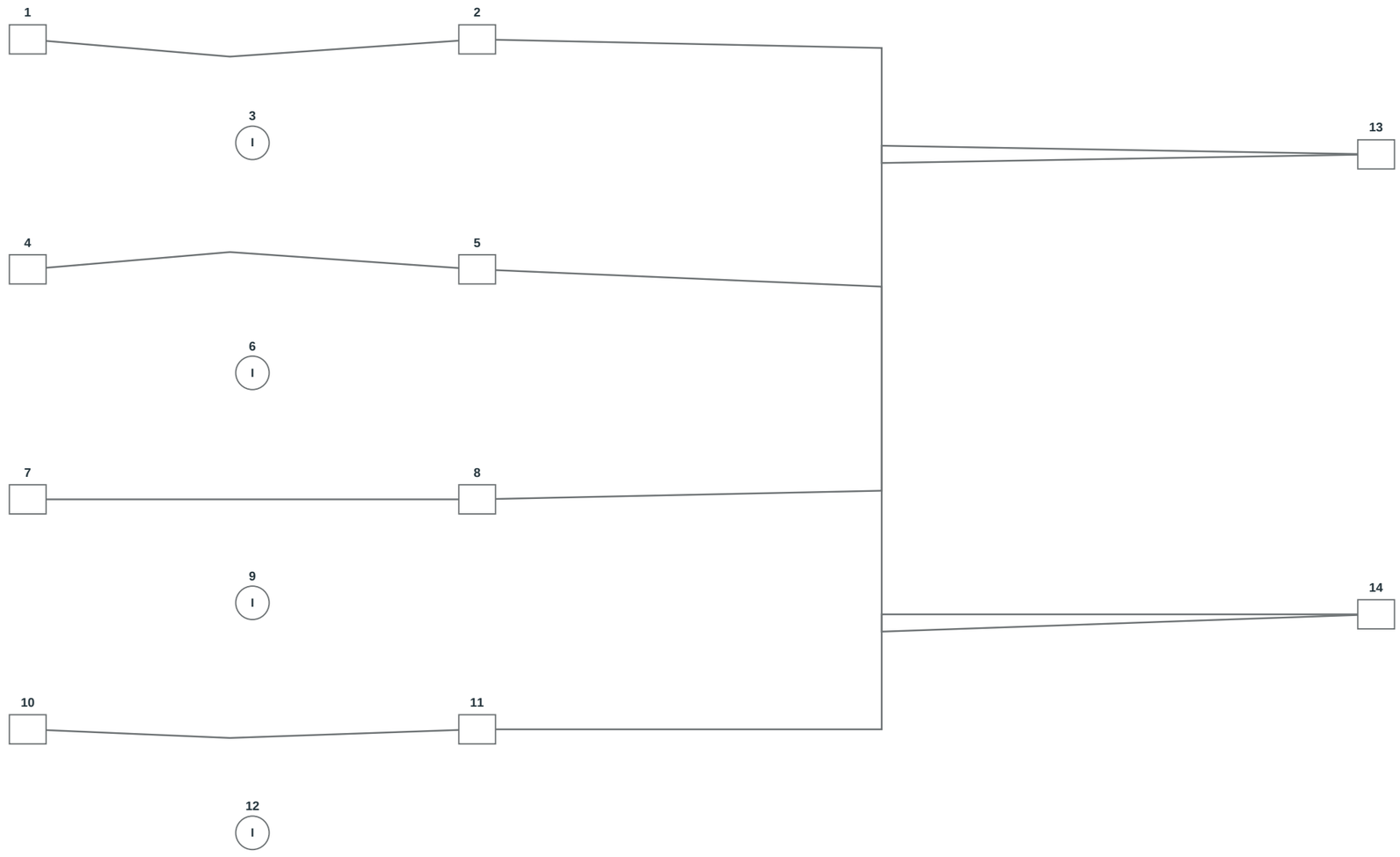
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 43

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Exhaust gas - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	tc-mainP	11XJB10 BR001
2	sil-mainP	11XJR10 BS001
3	temp-mainP	11XJR10 CT001
4	tc-mainS	12XJB10 BR001
5	sil-mainS	12XJR10 BS001
6	temp-mainS	12XJR10 CT001
7	tc-gen1	13XJB10 BR001
8	sil-gen1	13XJR10 BS001
9	temp-gen1	13XJR10 CT001
10	tc-gen2	14XJB10 BR001
11	sil-gen2	14XJR10 BS001
12	temp-gen2	14XJR10 CT001
13	funP	10XJR20 BR001
14	funS	10XJR20 BR002

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Exhaust gas · 400 deg C

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-10 | Exhaust gas - system connection schematic

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 44

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Exhaust gas - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	tc-mainP	11XJB10 BR001	Port main engine: turbocharger turbine	engine-room	MISSING
2	sil-mainP	11XJR10 BS001	Silencer and spark arrester	engine-room	MISSING
3	temp-mainP	11XJR10 CT001	Exhaust temperature (EXHAUST on the console)	engine-room	MISSING
4	tc-mainS	12XJB10 BR001	Starboard main engine: turbocharger turbine	engine-room	MISSING
5	sil-mainS	12XJR10 BS001	Silencer and spark arrester	engine-room	MISSING
6	temp-mainS	12XJR10 CT001	Exhaust temperature (EXHAUST on the console)	engine-room	MISSING
7	tc-gen1	13XJB10 BR001	Generator No.1 engine: turbocharger turbine	aux-room	MISSING
8	sil-gen1	13XJR10 BS001	Silencer and spark arrester	aux-room	MISSING
9	temp-gen1	13XJR10 CT001	Exhaust temperature (EXHAUST on the console)	aux-room	MISSING
10	tc-gen2	14XJB10 BR001	Generator No.2 engine: turbocharger turbine	aux-room	MISSING
11	sil-gen2	14XJR10 BS001	Silencer and spark arrester	aux-room	MISSING
12	temp-gen2	14XJR10 CT001	Exhaust temperature (EXHAUST on the console)	aux-room	MISSING
13	funP	10XJR20 BR001	Funnel P: port main engine and generator No.1	funnel-p	MISSING
14	funS	10XJR20 BR002	Funnel S: starboard main engine and generator No.2	funnel-s	MISSING

SYSTEM NOTES

1. Lagged with mineral wool under stainless cladding; no exhaust pipe within 0.6 m of a fuel line.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-10-N1 | Exhaust gas - equipment and node schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 45

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Exhaust gas - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
EXHAUST-001	tc-mainP	sil-mainP	200	MISSING	
EXHAUST-002	sil-mainP	funP	200	MISSING	lagged, up casing P
EXHAUST-003	tc-gen1	sil-gen1	125	MISSING	
EXHAUST-004	sil-gen1	funP	125	MISSING	
EXHAUST-005	tc-mainS	sil-mainS	200	MISSING	
EXHAUST-006	sil-mainS	funS	200	MISSING	lagged, up casing S
EXHAUST-007	tc-gen2	sil-gen2	125	MISSING	
EXHAUST-008	sil-gen2	funS	125	MISSING	

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-10-L1 | Exhaust gas - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

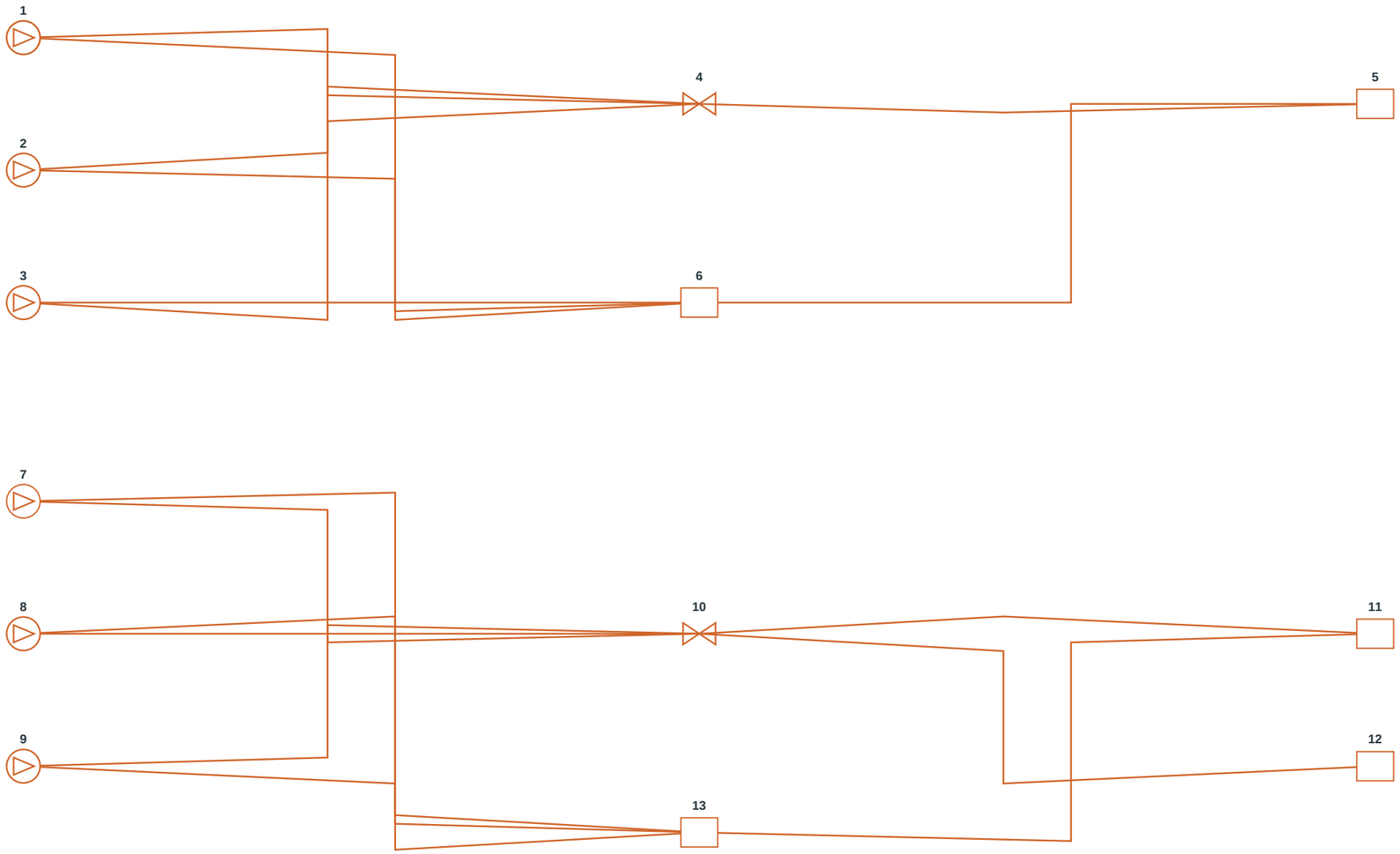
REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 46

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Hydraulics: steering and ramp - system connection schematic

Topology from ferry-piping.js; equipment nodes retain original tags. DN values require hydraulic verification.



No.	Node ID	Equipment tag
1	hpu1	10XSA10 AP001
2	hpu2	10XSA10 AP002
3	hand	10XSA10 AP003
4	hvalve	10XSA10 AA201
5	rams	10XSA10 BR001
6	htank	10XSA10 BB001
7	rp1	10XSB10 AP001
8	rp2	10XSB10 AP002
9	rhand	10XSB10 AP003
10	rvalve	10XSB10 AA202
11	cyl	10XSB10 BR001
12	pins	10XSB10 AA203
13	rtank	10XSB10 BB001

Symbols: circle/triangle pump; bow-tie valve; I instrument; rectangle equipment. Indicative symbols only.

Line direction and valve fail position are not implied. Complete IDs, DN and notes in piping_line_schedule.csv.

Medium: Hydraulic oil · 14 MPa

Release hold: hydraulic capacity / pipe class / isolation / instrumentation / fail-safe logic / material / approved maker interfaces.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-11 | Hydraulics: steering and ramp - system connection schem...

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 47

Hydraulics: steering and ramp - equipment and node schedule

Node tag and compartment authority: source module; maker approval and process calculations are open.

No.	Node	Tag	Equipment / function	Space	Port
1	hpu1	10XSA10 AP001	Steering power unit No.1 · 2.2 kW	steering-gear	MISSING
2	hpu2	10XSA10 AP002	Steering power unit No.2 · 2.2 kW · emergency board	steering-gear	MISSING
3	hand	10XSA10 AP003	Hand pump	steering-gear	MISSING
4	hvalve	10XSA10 AA201	Solenoid valve block (follow-up, non-follow-up)	steering-gear	MISSING
5	rams	10XSA10 BR001	Two rams on the tiller and tie bar · 35 deg	steering-gear	MISSING
6	htank	10XSA10 BB001	Oil tank · 60 L · level switch	steering-gear	MISSING
7	rp1	10XSB10 AP001	Ramp pump No.1 · 5.5 kW	bow-house-p	MISSING
8	rp2	10XSB10 AP002	Ramp pump No.2 · 5.5 kW	bow-house-p	MISSING
9	rhand	10XSB10 AP003	Hand pump	bow-house-p	MISSING
10	rvalve	10XSB10 AA202	Control valve block · up, down, hold	bow-house-p	MISSING
11	cyl	10XSB10 BR001	Two cylinders · 100 mm · 14 MPa	ramp	MISSING
12	pins	10XSB10 AA203	Two locking pins · limit switches to the bridge lamp	ramp	MISSING
13	rtank	10XSB10 BB001	Oil tank · 120 L	bow-house-p	MISSING

SYSTEM NOTES

1. Electro-hydraulic, two power units: one at sea, both in narrow water; No.2 is fed from the emergency switchboard; a hand pump and a telephone to the bridge.

2. One pump runs and one stands by; a hand pump if both fail.

CONSTRUCTION RELEASE HOLD

1. Equipment labels and ratings are fictional unless explicitly listed in the reference particulars. Line sizing is an assumption pending flow, pressure drop and pump curve verification.

2. See full line schedule CSV for each branch and missing physical route. No approved bill of material or fabrication isometric is supplied.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-11-N1 | Hydraulics: steering and ramp - equipment and node sche...

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 48

Hydraulics: steering and ramp - line connection schedule

Each DN is a source nominal design value. Material, pressure class, valve counts and testing remain unverified.

Line ID	From node	To node	DN	Route	Note
HYDRAULICS-001	htank	hpu1	20	MISSING	
HYDRAULICS-002	htank	hpu2	20	MISSING	
HYDRAULICS-003	htank	hand	15	MISSING	
HYDRAULICS-004	hpu1	hvalve	15	MISSING	
HYDRAULICS-005	hpu2	hvalve	15	MISSING	
HYDRAULICS-006	hand	hvalve	15	MISSING	
HYDRAULICS-007	hvalve	rams	15	MISSING	
HYDRAULICS-008	rams	htank	20	MISSING	return
HYDRAULICS-009	rtank	rp1	25	MISSING	
HYDRAULICS-010	rtank	rp2	25	MISSING	
HYDRAULICS-011	rtank	rhand	15	MISSING	
HYDRAULICS-012	rp1	rvalve	20	MISSING	
HYDRAULICS-013	rp2	rvalve	20	MISSING	
HYDRAULICS-014	rhand	rvalve	15	MISSING	
HYDRAULICS-015	rvalve	cyl	20	MISSING	
HYDRAULICS-016	rvalve	pins	12	MISSING	
HYDRAULICS-017	cyl	rtank	25	MISSING	return

CONNECTION SCHEDULE QUALIFICATION

1. Signal links (DN0) are not pipe lines. Model routes terminate at game connection anchors; their coordinates are not vendor nozzle locations.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-PID-11-L1 | Hydraulics: steering and ramp - line connection schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 49

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Electrical distribution - corrected one-line

Primary and secondary sides shown separately; breaker frames and provisional settings in A. No fault/selectivity approval.

DG1

125 kVA | 445 V nameplate | 60 Hz

200 frame / 170 setting

DG2

125 kVA | 445 V nameplate | 60 Hz

200 frame / 170 setting

SHORE TRANSFORMER

200 -> 440 V | 50 kVA | 60 Hz required

P: 144.3 A | 160 frame / 145 setting

S: 65.6 A | 80 frame / 66 setting

No generator parallel

EG - 40 kVA

445 V nameplate | 440 V nominal | 32 kW

63 frame / 52 setting

MSB - 440 V nominal

IT system | synchronisation and generator protection required

P: 39.4 A | 50 frame / 40 setting

MSB / EG changeover interlock

ESB - 440 V

Independent emergency source; MSB tie interlocked

T1 / T2 - 30 kVA

440 ->105 V | 3 phase | one duty, one standby

LB - 105 V

S: 165.0 A | 200 frame / 165 setting

Interlocked selector; no parallel

ETR - 5 kVA

440 ->105 V | emergency lighting | I 6.6 /27.5 A

P: 6.6 A | 10 frame / 7 setting

S: 27.5 A | 32 frame / 28 setting

DESIGN HOLDS

1. Generator bus voltage regulation, feeder source impedance, transformer connection groups, frequency compatibility, starting transient, short-circuit duty, selectivity and emergency endurance need verification.

2. PE is separate from loaded conductors. Star-delta motors require six motor conductors plus PE from starter to motor.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-EL-001 | Electrical distribution - corrected one-line

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 50

Sources and transformer side schedule

Source current and protection are side-specific; loaded conductor count excludes PE.

Source	Side	Voltage	Rated A	Frame A	Setting A	Cable
DG1	output	445 V	162.2	200	170	2 parallel x (3 x 95 mm2) + dedicated PE 95 mm2
DG2	output	445 V	162.2	200	170	2 parallel x (3 x 95 mm2) + dedicated PE 95 mm2
EG	output	445 V	51.9	63	52	3 x 16 mm2 + PE 16 mm2
T1	primary	440 V	39.4	50	40	3 x 10 mm2 + PE 10 mm2
T1	secondary	105 V	165.0	200	165	3 x 95 mm2 + PE 95 mm2
T2	primary	440 V	39.4	50	40	3 x 10 mm2 + PE 10 mm2
T2	secondary	105 V	165.0	200	165	3 x 95 mm2 + PE 95 mm2
ETR	primary	440 V	6.6	10	7	3 x 1.5 mm2 + PE 1.5 mm2
ETR	secondary	105 V	27.5	32	28	3 x 6 mm2 + PE 6 mm2
shore	primary	200 V	144.3	160	145	3 x 70 mm2 + PE 70 mm2
shore	secondary	440 V	65.6	80	66	3 x 25 mm2 + PE 25 mm2

CALCULATION BASIS

1. Three-phase I=S/(sqrt(3)U). Lighting transformer30 kVA: 39.36 A at 440 V; 164.96 A at 105 V. Shore50 kVA: 144.34 A at 200 V; 65.61 A at 440 V.
2. Primary and secondary conductors must be independently protected and sized. A 50 A primary breaker does not establish adequate 105 V secondary cable protection.
3. Current game cable ampacity assumptions are not verified vendor tables. Apply actual cable construction, ambient, grouping, thermal and short-circuit withstand factors before release.
4. Full cable and feeder metadata, including PE and motor-tail conductors where available, are exported in electrical_feeders.csv.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-EL-002 | Sources and transformer side schedule

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 51

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Feeder and cable schedule - 1

Preliminary cable selection; voltage-drop correction includes copper conductor operating temperature.

ID	Board	Load / consumer	A	CB A	Cable + PE	L m	Drop %
steer-1	MSB	Steering gear pump No.1 · overload alarm, no ...	4	6	3 x 1.5 mm2 + PE 1.5 mm2	15	0.4
steer-2	ESB	Steering gear pump No.2 · overload alarm, no ...	4	6	3 x 1.5 mm2 + PE 1.5 mm2	25	0.6
fire-1	MSB	Fire and general-service pump No.1 (star-delta)	19.1	25	3 x 4 mm2 + PE 4 mm2	7	0.3
fire-2	MSB	Fire and bilge pump No.2 (star-delta)	19.1	25	3 x 4 mm2 + PE 4 mm2	7	0.3
bilge	MSB	Bilge pump	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	7	0.3
fire-emergency	ESB	Emergency fire pump	10.1	16	3 x 4 mm2 + PE 4 mm2	25	0.6
er-fansA	MSB	Engine-room supply fans feeder 1	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	16	0.6
er-fansB	MSB	Engine-room supply fans feeder 2	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	16	0.6
car-deck-fansA	MSB	Car-deck exhaust fans feeder 1	4	6	3 x 1.5 mm2 + PE 1.5 mm2	9	0.2
car-deck-fansB	MSB	Car-deck exhaust fans feeder 2	4	6	3 x 1.5 mm2 + PE 1.5 mm2	9	0.2
thruster	MSB	Bow thruster (star-delta, two-generator inter...	128.6	200	3 x 95 mm2 + PE 95 mm2	28	0.3
windlass	MSB	Anchor windlass and warping heads	13	20	3 x 6 mm2 + PE 6 mm2	39	0.7
capstan	MSB	Aft mooring capstan	10.1	16	3 x 2.5 mm2 + PE 2.5 mm2	20	0.7
rampA	MSB	Ramp power pack pump (one of two runs) feeder 1	10.1	16	3 x 6 mm2 + PE 6 mm2	39	0.6
rampB	MSB	Ramp power pack pump (one of two runs) feeder 2	10.1	16	3 x 6 mm2 + PE 6 mm2	39	0.6
compressor	MSB	Starting-air compressor No.1	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	7	0.3
compressor-2	MSB	Starting-air compressor No.2	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	13	0.5
fuel-transfer	MSB	Fuel transfer pump No.1	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.2
fuel-transfer-2	MSB	Fuel transfer pump No.2	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.2
purifier	MSB	Fuel purifier	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0.1

VERIFICATION LIMIT

1. Numeric checks cover stated ampacity and steady voltage drop only. Motor-start terminal voltage also depends on source transient response; breaker settings, discrimination, short-circuit ratings and final routes are not approved.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-EL-F01 | Feeder and cable schedule - 1

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 52

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Feeder and cable schedule - 2

Preliminary cable selection; voltage-drop correction includes copper conductor operating temperature.

ID	Board	Load / consumer	A	CB A	Cable + PE	L m	Drop %
lube-primingA	MSB	Main-engine lubricating-oil priming pumps fee...	1.5	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0.1
lube-primingB	MSB	Main-engine lubricating-oil priming pumps fee...	1.5	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0.1
hydrophore	MSB	Fresh-water hydrophore pump No.1	1.5	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0.1
hydrophore-2	MSB	Fresh-water hydrophore pump No.2	1.5	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.1
sanitary	MSB	Sanitary (sea-water flushing) pump No.1	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0.1
sanitary-2	MSB	Sanitary pump No.2	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.2
sewage	MSB	Sewage discharge pump No.1	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.2
sewage-2	MSB	Sewage discharge pump No.2	3.1	6	3 x 1.5 mm2 + PE 1.5 mm2	13	0.2
separator	MSB	Oily-water separator	0.8	6	3 x 1.5 mm2 + PE 1.5 mm2	7	0
davit	MSB	Rescue-boat davit winch	6.8	10	3 x 2.5 mm2 + PE 2.5 mm2	23	0.5
saloon-acA	MSB	Saloon package air conditioners feeder 1	13	16	3 x 1.5 mm2 + PE 1.5 mm2	26	2
saloon-acB	MSB	Saloon package air conditioners feeder 2	13	16	3 x 1.5 mm2 + PE 1.5 mm2	26	2
range	MSB	Galley range	7.9	10	3 x 1.5 mm2 + PE 1.5 mm2	24	1.1
water-heater	MSB	Water heater, 100 L	3.9	6	3 x 1.5 mm2 + PE 1.5 mm2	24	0.5
chargersA	MSB	24 V battery chargers feeder 1	2	6	3 x 1.5 mm2 + PE 1.5 mm2	24	0.3
chargersB	MSB	24 V battery chargers feeder 2	2	6	3 x 1.5 mm2 + PE 1.5 mm2	24	0.3
workshop	MSB	Workshop lathe, drill and grinder (3 units)	2.4	6	3 x 1.5 mm2 + PE 1.5 mm2	20	0.3
ballast	MSB	Ballast pump	6.8	10	3 x 1.5 mm2 + PE 1.5 mm2	13	0.5
light-passenger	LB	Lighting: saloon, carpet room, lobby, toilets...	25	25	2 x 6 mm2 + PE 6 mm2	26	4.6
light-car-deck	LB	Lighting: car deck	10.2	16	2 x 2.5 mm2 + PE 2.5 mm2	24	4.2

VERIFICATION LIMIT

1. Numeric checks cover stated ampacity and steady voltage drop only. Motor-start terminal voltage also depends on source transient response; breaker settings, discrimination, short-circuit ratings and final routes are not approved.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-EL-F02 | Feeder and cable schedule - 2

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 53

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Feeder and cable schedule - 3

Preliminary cable selection; voltage-drop correction includes copper conductor operating temperature.

ID	Board	Load / consumer	A	CB A	Cable + PE	L m	Drop %
light-engine	LB	Lighting: engine room, steering-gear room	11.2	16	2 x 1.5 mm2 + PE 1.5 mm2	7	2.2
light-crew	LB	Lighting: bridge, mess, galley, cabins	4.4	6	2 x 1.5 mm2 + PE 1.5 mm2	28	3.5
floodlights	LB	Deck lights and ramp floodlights (6 x 100 W...	27.5	32	2 x 6 mm2 + PE 6 mm2	24	4.7
searchlight	LB	Searchlight	9.5	10	2 x 4 mm2 + PE 4 mm2	37	3.7
bridge-electronics	LB	Bridge electronics (radar, autopilot, echo so...	7.9	10	2 x 2.5 mm2 + PE 2.5 mm2	33	4.4
room-acA	LB	Room air conditioners (bridge, mess, cabins) ...	22.4	25	2 x 6 mm2 + PE 6 mm2	28	4.4
room-acB	LB	Room air conditioners (bridge, mess, cabins) ...	22.4	25	2 x 6 mm2 + PE 6 mm2	28	4.4
galley-appliances	LB	Galley appliances (rice cooker, microwave, po...	38.1	40	2 x 10 mm2 + PE 10 mm2	24	3.9
kiosk	LB	Kiosk (ice-cream freezer, showcase, hot water)	18.1	20	2 x 4 mm2 + PE 4 mm2	17	3.3
vendingA	LB	Drink vending machines feeder 1	6.7	10	2 x 1.5 mm2 + PE 1.5 mm2	17	3.2
vendingB	LB	Drink vending machines feeder 2	6.7	10	2 x 1.5 mm2 + PE 1.5 mm2	17	3.2
tv	LB	Carpet-room television	0.8	6	2 x 1.5 mm2 + PE 1.5 mm2	20	0.5
toilet-fans	LB	Toilet exhaust fans (2 units)	3.4	6	2 x 1.5 mm2 + PE 1.5 mm2	16	1.5
lane-signals	LB	Lane signals at the ramp (2 posts, a board ea...	1.9	6	2 x 1.5 mm2 + PE 1.5 mm2	24	1.3
emergency-light	ESB	Emergency lighting (one fitting in four, the ...	16.4	20	2 x 2.5 mm2 + PE 2.5 mm2	10	2.8
nav-lights	ESB	Navigation lights (7 x 60 W)	4.4	6	2 x 1.5 mm2 + PE 1.5 mm2	24	3
emergency-radar	ESB	Radar and PA on the emergency supply	5.3	6	2 x 1.5 mm2 + PE 1.5 mm2	20	3
emergency-charger	ESB	24 V emergency charger (radio, fire detection...	4.8	6	3 x 1.5 mm2 + PE 1.5 mm2	2	0.1

VERIFICATION LIMIT

1. Numeric checks cover stated ampacity and steady voltage drop only. Motor-start terminal voltage also depends on source transient response; breaker settings, discrimination, short-circuit ratings and final routes are not approved.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-EL-F03 | Feeder and cable schedule - 3

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 54

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

Drawing production and validation record

Validation scope is explicitly bounded; no claim of shipbuilding approval or construction completeness.

Check	Evidence	Disposition
Geometry scale	Fixed mm geometry ->1:50 PDF transform	Exact transform; print calibration required
Paper size	A2 594 x 420 /A3 420 x 297 mm	All pages set explicitly
CAD editable units	DXF R2010; INSUNITS4	mm geometry in model space
Coordinate datums	Game-to-CAD transform in source snapshot	Transom /CL /BL
Node connectivity	Every source P&ID node and edge exported	Model topology; incomplete physical routing
Electrical side values	Transformer primary /secondary fields	Corrected preliminary calculations
Machinery source	Fictional game source +Yanmar reference	No approved vendor installation
Maintenance	Proposed machine/removal envelope drawing	Not supplier-verified; overhead limited
Construction release	Class/flag/maker/lift/foundation hold items	NOT RELEASED
Drawn/checked/approved	Generated; independent computational review	No real professional approval claimed

FINAL REVIEW INSTRUCTIONS

1. Compare source snapshot hashes with the repository engineering modifications and rerun the generator after any dimensional or system change.
2. Reopen every DXF and validate units and drawing register. Check PDF page bounds, title blocks, scale bars and rendered legibility.
3. Review source piping corrections, the bilge/ballast separation, actual emergency suction connection and overboard controls before any operational use.
4. Do not call this package construction-ready. Procurement and yard release require selected makers, signed calculations, approved drawings and applicable surveys.

Source e87c06b16734 | CAD units mm | dimensions govern

Print at 100%; verify calibration bar before measuring.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

JS127-ER-900 | Drawing production and validation record

DRAWN: generated design | CHECK: independent computational review | APPROVAL: NONE

REV B | 2026-10-13 | A3 landscape | SCALE NTS | SHEET 55